



PREMIER UNIVERSITY CHATTOGRAM

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT REPORT

COURSE NAME	Big Data Analysis Laboratory	
COURSE CODE	CSE 4346	
PROJECT TITLE	Steam Game Recommendations Data Analysis using Hadoop	
DATE OF REPORT	08-09-2026	
DATE OF SUBMISSION	15-09-2026	
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1. Problem Statement

Analyse the Steam game-recommendations dataset with Apache Hadoop (MapReduce) and Apache Pig, across its game and review tables, to find which games and rating groups draw the most reviews and the most hours of play on the Steam platform.

The work is split the way the course asks for it:

- **Three MapReduce programs in Java**, each built on a different MapReduce design pattern — summarization, data organization (binning) and a reduce-side join.
- **Five Apache Pig Latin scripts** that ask ranking and grouping questions of the same data, including a map-side replicated join over the full 41-million-row review table.

Everything was run on a single-node Hadoop cluster, and every job result in this report was read back out of HDFS after the job that produced it had finished.

2. Dataset Description

Dataset: Game Recommendations on Steam (Kaggle)

<https://www.kaggle.com/datasets/antonkozyriev/game-recommendations-on-steam>

The dataset is a crawl of the Steam store and its user reviews. It comes as three related CSV tables and one JSON file of game descriptions. `app_id` links a review to its game and `user_id` links it to its author. The analyses below use `games.csv` and `recommendations.csv`; `users.csv` is described for completeness.

File	Size	Rows	One row is
<code>recommendations.csv</code>	2.0 GB	41,154,794	one user review of one game
<code>users.csv</code>	193 MB	14,306,064	one Steam user
<code>games.csv</code>	4.9 MB	50,872	one game on the Steam store
<code>games_metadata.json</code>	18 MB	50,872	description and tags of a game (not used)

Table 1: Files in the dataset (row counts exclude the header line).

All three CSV files are comma separated, start with a header row and contain no quoted fields. Before any job was written I checked that every row has the expected number of columns (13 for games, 3 for users, 8 for recommendations); none failed, so the files needed no cleaning step beyond skipping the header line.

games.csv

#	Field	Type	Example	Meaning
0	app_id	int	13500	Steam application id (key)
1	title	string	Prince of Persia: Warrior Within™	game title
2	date_release	date	2008-11-21	release date
3–5	win, mac, linux	boolean	true, false, false	supported platforms
6	rating	string	Very Positive	Steam's review summary label (9 values)
7	positive_ratio	int	84	% of Steam reviews that are positive
8	user_reviews	int	2199	number of reviews on Steam
9	price_final	float	9.99	price in USD after discount
10	price_original	float	9.99	price in USD before discount
11	discount	float	0.0	discount in %
12	steam_deck	boolean	true	runs on the Steam Deck

users.csv

#	Field	Type	Meaning
0	user_id	int	anonymised user id (key)
1	products	int	number of games the user owns
2	reviews	int	number of reviews the user has written

recommendations.csv

#	Field	Type	Example	Meaning
0	app_id	int	975370	game reviewed → games.app_id
1	helpful	int	0	users who marked the review helpful
2	funny	int	0	users who marked the review funny
3	date	date	2022-12-12	review date
4	is_recommended	boolean	true	thumbs up or thumbs down
5	hours	float	36.3	hours the reviewer had played the game
6	user_id	int	51580	reviewer → users.user_id
7	review_id	int	0	unique review id

The rating column is the label Steam shows on a store page. It takes nine values, and it is the category field used for grouping and binning throughout this project, the same role the video category played in the YouTube lab.

3. Summary of Data Analysis

The following MapReduce design patterns are implemented:

1. **Summarization** — numerical summarization with a custom Writable and a combiner.
2. **Data Organization** — binning with MultipleOutputs.
3. **Join Patterns** — reduce-side join with MultipleInputs.

The following MapReduce analyses are performed on the dataset:

1. **Review summary per game** — number of reviews, recommended share, total, average and maximum hours played, and helpful votes for every game (*summarization*).
2. **Binning games by rating** — every game written to one of nine output files, one per Steam rating label (*data organization*).
3. **Games joined with their reviews** — each game's title and rating put next to its review count and total hours played (*reduce-side join*).

The following Pig analyses are performed on the dataset:

1. Top 5 rating groups by number of games
2. Top 10 most reviewed games
3. Top 10 most reviewed games in each rating group
4. Top 10 games by total hours played
5. Top 10 games by total hours played in each rating group

4. Environment and Loading the Data

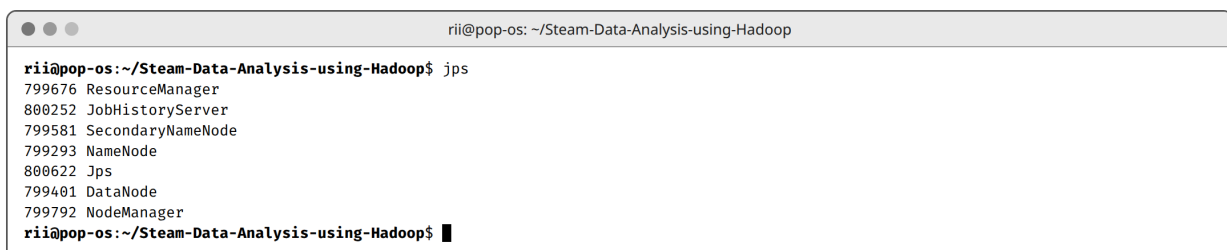
Component	Version / setting
Operating system	Pop!_OS Linux, 8 CPU cores, 16 GB RAM
Apache Hadoop	3.4.1, pseudo-distributed (single node), fs.defaultFS = hdfs://localhost:9000
HDFS	replication 1, block size 128 MB
Apache Pig	0.18.0, run in mapreduce mode on YARN
Java	OpenJDK 11.0.31
YARN memory	NodeManager 5120 MB; map 768 MB, reduce 1024 MB, AM 768 MB containers

Table 2: Software used.

The laptop had only about 2 GB of RAM free, and with Hadoop's default settings YARN would try to run eight 1 GB task containers at once. The NodeManager was therefore capped at 5 GB and the task containers made smaller in `yarn-site.xml` and `mapred-site.xml`, so the jobs run a few tasks at a time instead of pushing the machine into swap.

Starting the cluster

The cluster is started daemon by daemon, followed by the JobHistory server. `jps` shows all six Hadoop daemons running:



```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ jps
799676 ResourceManager
800252 JobHistoryServer
799581 SecondaryNameNode
799293 NameNode
800622 Jps
799401 DataNode
799792 NodeManager
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$

```

Figure 1: All six Hadoop daemons running (`jps`).

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ hadoop version | head -1; pig -version 2>&1 | grep -i 'Apache Pig'; java -version 2>&1
| head -1
Hadoop 3.4.1
Apache Pig version 0.18.0 (r1926484)
openjdk version "11.0.31" 2026-04-21
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ █

```

Figure 2: Hadoop, Pig and Java versions.

Loading the dataset into HDFS

The three CSV files are copied into the /steam directory in HDFS. The 2 GB review file is split into 16 blocks of 128 MB, and `hdfs fsck` reports the directory as healthy.

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ hdfs dfs -mkdir -p /steam && cd dataset && time hdfs dfs -put -f games.csv users.csv r
ecommendations.csv /steam/
2026-09-11 20:15:27,306 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
2026-09-11 20:15:28,552 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable

real    0m11.513s
user    0m8.256s
sys     0m4.015s
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ █

```

Figure 3: Copying the dataset into HDFS (11.5 seconds).

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ hdfs dfs -ls -h /steam && hdfs fsck /steam -files -blocks | grep -E 'recommendations.c
sv|Total blocks|Status'
2026-09-11 20:15:40,134 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
Found 3 items
-rw-r--r--  1 rii supergroup      4.6 M 2026-09-11 20:15 /steam/games.csv
-rw-r--r--  1 rii supergroup      1.9 G 2026-09-11 20:15 /steam/recommendations.csv
-rw-r--r--  1 rii supergroup    184.1 M 2026-09-11 20:15 /steam/users.csv
2026-09-11 20:15:41,784 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
Connecting to namenode via http://localhost:9870/fsc?ugi=rii&files=1&blocks=1&path=%2Fsteam
/steam/recommendations.csv 2023223067 bytes, replicated: replication=1, 16 block(s): OK
Status: HEALTHY
Total blocks (validated):      19 (avg. block size 116902888 B)
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ █

```

Figure 4: The dataset in HDFS: 1.9 GB of reviews stored as 16 blocks.

Building the MapReduce jobs

Each job is compiled against the local Hadoop classpath and packed into its own jar by `scripts/build.sh`:

```

1 #!/usr/bin/env bash
2 # Compile each MapReduce job against the local Hadoop and package it as
3 # <Job>/<Job>.jar. Needs 'hadoop' on PATH (Hadoop 3.4.1, Java 11).
4 set -euo pipefail
5 cd "$(dirname "$0")/.."
6
7 for job in GameReviewSummary BinningByRating GameReviewJoin; do
8     rm -rf "build/$job" && mkdir -p "build/$job"
9     javac -cp "$(hadoop classpath)" -d "build/$job" $(find "$job/src/main/java" -name '*.java')
10    jar cf "$job/$job.jar" -C "build/$job" .
11    echo "built $job/$job.jar"
12 done

```

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ ./scripts/build.sh && ls -l */*.jar
built GameReviewSummary/GameReviewSummary.jar
built BinningByRating/BinningByRating.jar
built GameReviewJoin/GameReviewJoin.jar
-rw-rw-r-- 1 rii rii 2849 Sep 11 20:18 BinningByRating/BinningByRating.jar
-rw-rw-r-- 1 rii rii 6076 Sep 11 20:18 GameReviewJoin/GameReviewJoin.jar
-rw-rw-r-- 1 rii rii 4996 Sep 11 20:18 GameReviewSummary/GameReviewSummary.jar
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ █

```

Figure 5: The three jars built.

5. Map Reduce Analysis

5.1 Review Summary per Game

Returns, for every game, how many reviews it received, how many recommend it, how long its reviewers played it and how many helpful votes its reviews collected (GameReviewSummary).

This is the **numerical summarization** pattern. The mapper reads one review at a time and emits the game's app_id with a small tuple holding a count of one, a 1 or 0 for “recommended”, the hours played and the helpful votes. Every field of the tuple is additive — a count, a sum or a maximum — so partial results can be merged in any order. That is what makes a **combiner** valid here: each mapper merges its own tuples before the shuffle, and the reducer finishes the job with exactly the same code.

This analysis consists of:

1. **Driver Class** — sets up the job, and registers the reducer as the combiner as well.
2. **Mapper Class** — emits app_id of type Text and a tuple of custom type ReviewStatsTuple.
3. **Reducer Class** — merges all tuples of one game and emits app_id with the final ReviewStatsTuple; also used as the combiner.
4. **Tuple Class** — custom Writable holding reviews, recommended, total hours, maximum hours and helpful votes.

Driver Class

```

1 package steam.summary;
2
3 import org.apache.hadoop.conf.Configuration;
4 import org.apache.hadoop.fs.Path;
5 import org.apache.hadoop.io.Text;
6 import org.apache.hadoop.mapreduce.Job;
7 import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
8 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
9
10 /**
11  * MR-1 Review summary per game (numerical summarization pattern).
12  *
13  * For every app_id in recommendations.csv: number of reviews, number
14  * recommended, positive %, total / average / maximum hours played and total
15  * helpful votes.
16  *
17  * Usage: hadoop jar GameReviewSummary.jar steam.summary.GameReviewSummary <in> <out>
18  */
19 public class GameReviewSummary {
20
21     public static void main(String[] args) throws Exception {
22         if (args.length != 2) {

```

```

23         System.err.println("Usage: GameReviewSummary <input> <output>");
24         System.exit(2);
25     }
26     Configuration conf = new Configuration();
27     Job job = Job.getInstance(conf, "Steam review summary per game");
28     job.setJarByClass(GameReviewSummary.class);
29
30     job.setMapperClass(ReviewMapper.class);
31     job.setCombinerClass(ReviewReducer.class);
32     job.setReducerClass(ReviewReducer.class);
33
34     job.setOutputKeyClass(Text.class);
35     job.setOutputValueClass(ReviewStatsTuple.class);
36
37     FileInputFormat.addInputPath(job, new Path(args[0]));
38     FileOutputFormat.setOutputPath(job, new Path(args[1]));
39     System.exit(job.waitForCompletion(true) ? 0 : 1);
40 }
41 }

```

Mapper Class

```

1  package steam.summary;
2
3  import java.io.IOException;
4
5  import org.apache.hadoop.io.LongWritable;
6  import org.apache.hadoop.io.Text;
7  import org.apache.hadoop.mapreduce.Mapper;
8
9  /**
10   * Reads recommendations.csv
11   * (app_id, helpful, funny, date, is_recommended, hours, user_id, review_id)
12   * and emits (app_id, tuple) for every review, with a count of one.
13   */
14  public class ReviewMapper extends Mapper<LongWritable, Text, Text, ReviewStatsTuple> {
15
16      private final Text appId = new Text();
17      private final ReviewStatsTuple tuple = new ReviewStatsTuple();
18
19      @Override
20      protected void map(LongWritable key, Text value, Context context)
21          throws IOException, InterruptedException {
22          String[] f = value.toString().split(",");
23          if (f.length != 8 || f[0].equals("app_id")) {
24              return; // header or malformed row
25          }
26          try {
27              long helpful = Long.parseLong(f[1]);
28              boolean recommended = Boolean.parseBoolean(f[4]);
29              double hours = Double.parseDouble(f[5]);
30              appId.set(f[0]);
31              tuple.set(1, recommended ? 1 : 0, hours, hours, helpful);
32              context.write(appId, tuple);
33          } catch (NumberFormatException e) {
34              context.getCounter("Summary", "Malformed rows").increment(1);
35          }
36      }
37  }

```

Reducer Class (also the Combiner)

```

1 package steam.summary;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.io.Text;
6 import org.apache.hadoop.mapreduce.Reducer;
7
8 /**
9  * Merges every tuple for one app_id into a single summary.
10  * Input and output types match, so the driver also registers this class as
11  * the combiner: each mapper pre-merges its own tuples before the shuffle.
12  */
13 public class ReviewReducer extends Reducer<Text, ReviewStatsTuple, Text, ReviewStatsTuple> {
14
15     private final ReviewStatsTuple result = new ReviewStatsTuple();
16
17     @Override
18     protected void reduce(Text key, Iterable<ReviewStatsTuple> values, Context context)
19         throws IOException, InterruptedException {
20         result.set(0, 0, 0, 0, 0);
21         for (ReviewStatsTuple t : values) {
22             result.merge(t);
23         }
24         context.write(key, result);
25     }
26 }

```

Tuple Class

```

1 package steam.summary;
2
3 import java.io.DataInput;
4 import java.io.DataOutput;
5 import java.io.IOException;
6
7 import org.apache.hadoop.io.Writable;
8
9 /**
10  * Custom Writable that carries the running totals for one game.
11  * Every field is additive (count, sum, max), so the same tuple can be merged
12  * in the combiner and again in the reducer without losing accuracy.
13  */
14 public class ReviewStatsTuple implements Writable {
15
16     private long reviews;        // number of recommendations
17     private long recommended;    // how many of them were thumbs up
18     private double totalHours;    // hours played, summed over reviewers
19     private double maxHours;      // most hours any single reviewer played
20     private long helpful;        // "helpful" votes, summed
21
22     public ReviewStatsTuple() {
23     }
24
25     public void set(long reviews, long recommended, double totalHours, double maxHours, long helpful) {
26         this.reviews = reviews;
27         this.recommended = recommended;
28         this.totalHours = totalHours;
29         this.maxHours = maxHours;
30         this.helpful = helpful;
31     }
32
33     /** Folds another partial result into this one. */

```

```
34 public void merge(ReviewStatsTuple other) {
35     reviews += other.reviews;
36     recommended += other.recommended;
37     totalHours += other.totalHours;
38     maxHours = Math.max(maxHours, other.maxHours);
39     helpful += other.helpful;
40 }
41
42 public void write(DataOutput out) throws IOException {
43     out.writeLong(reviews);
44     out.writeLong(recommended);
45     out.writeDouble(totalHours);
46     out.writeDouble(maxHours);
47     out.writeLong(helpful);
48 }
49
50 public void readFields(DataInput in) throws IOException {
51     reviews = in.readLong();
52     recommended = in.readLong();
53     totalHours = in.readDouble();
54     maxHours = in.readDouble();
55     helpful = in.readLong();
56 }
57
58 /** reviews, recommended, positive %, total hours, average hours, max hours, helpful votes */
59 @Override
60 public String toString() {
61     double positivePct = reviews == 0 ? 0 : 100.0 * recommended / reviews;
62     double avgHours = reviews == 0 ? 0 : totalHours / reviews;
63     return reviews + "\t" + recommended + "\t" + String.format("%.1f", positivePct) + "\t"
64         + String.format("%.1f", totalHours) + "\t" + String.format("%.1f", avgHours) + "\t"
65         + String.format("%.1f", maxHours) + "\t" + helpful;
66 }
67 }
```

Execution

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ time hadoop jar GameReviewSummary/GameReviewSummary.jar steam.summary.GameReviewSummary /steam/recommendations.csv /steam/out_summary
2026-09-11 20:18:44,602 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-11 20:18:45,028 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:18:45,375 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-11 20:18:45,400 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/rii/.staging/job_1789136053647_0002
2026-09-11 20:18:45,697 INFO input.FileInputFormat: Total input files to process : 1
2026-09-11 20:18:46,584 INFO mapreduce.JobSubmitter: number of splits:15
2026-09-11 20:18:47,370 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789136053647_0002
2026-09-11 20:18:47,370 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-11 20:18:47,593 INFO conf.Configuration: resource-types.xml not found
2026-09-11 20:18:47,593 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-11 20:18:47,676 INFO impl.YarnClientImpl: Submitted application application_1789136053647_0002
2026-09-11 20:18:47,742 INFO mapreduce.Job: The url to track the job: http://pop-os:8088/proxy/application_1789136053647_0002/
2026-09-11 20:18:47,743 INFO mapreduce.Job: Running job: job_1789136053647_0002
2026-09-11 20:18:54,915 INFO mapreduce.Job: Job job_1789136053647_0002 running in uber mode : false
2026-09-11 20:18:54,916 INFO mapreduce.Job: map 0% reduce 0%
2026-09-11 20:19:20,019 INFO mapreduce.Job: map 12% reduce 0%
2026-09-11 20:19:26,294 INFO mapreduce.Job: map 18% reduce 0%
2026-09-11 20:19:27,302 INFO mapreduce.Job: map 20% reduce 0%
2026-09-11 20:19:28,604 INFO mapreduce.Job: map 22% reduce 0%
2026-09-11 20:19:32,739 INFO mapreduce.Job: map 25% reduce 0%
2026-09-11 20:19:33,761 INFO mapreduce.Job: map 27% reduce 0%
2026-09-11 20:19:34,847 INFO mapreduce.Job: map 31% reduce 0%
2026-09-11 20:19:35,895 INFO mapreduce.Job: map 33% reduce 0%
2026-09-11 20:19:49,273 INFO mapreduce.Job: map 36% reduce 0%
2026-09-11 20:19:55,040 INFO mapreduce.Job: map 39% reduce 0%
2026-09-11 20:19:58,104 INFO mapreduce.Job: map 45% reduce 0%
2026-09-11 20:19:59,122 INFO mapreduce.Job: map 45% reduce 11%
2026-09-11 20:20:00,154 INFO mapreduce.Job: map 48% reduce 11%
2026-09-11 20:20:01,220 INFO mapreduce.Job: map 49% reduce 11%
2026-09-11 20:20:04,311 INFO mapreduce.Job: map 52% reduce 11%
2026-09-11 20:20:05,323 INFO mapreduce.Job: map 52% reduce 13%
2026-09-11 20:20:06,338 INFO mapreduce.Job: map 58% reduce 13%
2026-09-11 20:20:07,388 INFO mapreduce.Job: map 60% reduce 13%
2026-09-11 20:20:11,462 INFO mapreduce.Job: map 60% reduce 20%
2026-09-11 20:20:20,732 INFO mapreduce.Job: map 63% reduce 20%
2026-09-11 20:20:26,937 INFO mapreduce.Job: map 64% reduce 20%
2026-09-11 20:20:27,959 INFO mapreduce.Job: map 67% reduce 20%
2026-09-11 20:20:29,112 INFO mapreduce.Job: map 73% reduce 20%
2026-09-11 20:20:32,484 INFO mapreduce.Job: map 75% reduce 20%
2026-09-11 20:20:33,508 INFO mapreduce.Job: map 77% reduce 20%
2026-09-11 20:20:34,515 INFO mapreduce.Job: map 84% reduce 20%
2026-09-11 20:20:36,543 INFO mapreduce.Job: map 87% reduce 20%
2026-09-11 20:20:38,567 INFO mapreduce.Job: map 87% reduce 29%
2026-09-11 20:20:51,831 INFO mapreduce.Job: map 91% reduce 29%
2026-09-11 20:20:53,861 INFO mapreduce.Job: map 96% reduce 29%
2026-09-11 20:20:54,880 INFO mapreduce.Job: map 98% reduce 29%
2026-09-11 20:20:55,886 INFO mapreduce.Job: map 100% reduce 29%
2026-09-11 20:20:57,896 INFO mapreduce.Job: map 100% reduce 100%
2026-09-11 20:20:58,913 INFO mapreduce.Job: Job job_1789136053647_0002 completed successfully
[... 66 more lines ...]

```

Figure 6: Review summary job running over the 2 GB review file.

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ time hadoop jar GameReviewSummary/GameReviewSummary.jar steam.summary.GameReviewSummary
/steam/recommendations.csv /steam/out_summary
[... 48 lines above ...]
2026-09-11 20:20:58,913 INFO mapreduce.Job: Job job_1789136053647_0002 completed successfully
2026-09-11 20:20:59,418 INFO mapreduce.Job: Counters: 55
  File System Counters
    FILE: Number of bytes read=12106480
    FILE: Number of bytes written=21542095
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=2023282091
    HDFS: Number of bytes written=1441145
    HDFS: Number of read operations=50
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Killed map tasks=1
    Launched map tasks=16
    Launched reduce tasks=1
    Data-local map tasks=16
    Total time spent by all maps in occupied slots (ms)=1400652
    Total time spent by all reduces in occupied slots (ms)=323000
    Total time spent by all map tasks (ms)=466884
    Total time spent by all reduce tasks (ms)=80750
    Total vcore-milliseconds taken by all map tasks=466884
    Total vcore-milliseconds taken by all reduce tasks=80750
    Total megabyte-milliseconds taken by all map tasks=358566912
    Total megabyte-milliseconds taken by all reduce tasks=82688000
  Map-Reduce Framework
    Map input records=41154795
    Map output records=41154794
    Map output bytes=1935187119
    Map output materialized bytes=4481013
    Input split bytes=1680
    Combine input records=41309286
    Combine output records=245258
    Reduce input groups=37610
    Reduce shuffle bytes=4481013
    Reduce input records=90766
    Reduce output records=37610
    Spilled Records=336024
    Shuffled Maps =15
    Failed Shuffles=0
    Merged Map outputs=15
    GC time elapsed (ms)=12253
    CPU time spent (ms)=384640
    Physical memory (bytes) snapshot=6200295424
    Virtual memory (bytes) snapshot=40950755328
    Total committed heap usage (bytes)=4475322368
    Peak Map Physical memory (bytes)=451526656
    Peak Map Virtual memory (bytes)=2555727872
    Peak Reduce Physical memory (bytes)=278171648
    Peak Reduce Virtual memory (bytes)=2810089472
  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
  File Input Format Counters
    Bytes Read=2023280411
  File Output Format Counters
    Bytes Written=1441145

real    2m15.426s
user    0m7.854s
sys     0m0.596s
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ █

```

Figure 7: Job counters of the review summary job.

The job read all 41,154,794 reviews in 15 input splits and finished in 2 minutes 15 seconds. The counters show the combiner at work: the mappers produced 41,154,794 records, but after combining only 90,766 of them reached the reducer — roughly 450 times less data through the shuffle. The reducer wrote one line for each of the **37,610 games** that have at least one review.

Output

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ hdfs dfs -ls /steam/out_summary 56 hdfs dfs -cat /steam/out_summary/part-r-000000 | sort -t$'\t' -k2 -nr | head -10
2026-09-11 20:21:28,716 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 1 rii supergroup 0 2026-09-11 20:20 /steam/out_summary/_SUCCESS
-rw-r--r-- 1 rii supergroup 1441145 2026-09-11 20:20 /steam/out_summary/part-r-000000
2026-09-11 20:21:29,853 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
440 319492 294879 92.3 101809977.6 318.7 999.9 521978
252490 270684 226293 83.6 92944004.6 343.4 999.9 771227
1091500 226414 168631 74.5 23146491.9 102.2 999.1 1189869
730 219737 186306 84.8 94260307.3 429.0 999.9 1910071
570 216914 171841 79.2 93072785.4 429.1 999.9 842346
444090 204176 171516 84.0 28549772.6 139.8 999.9 188983
292030 204166 193139 94.6 31522739.6 154.4 999.9 813793
377160 196789 160497 81.6 40149411.3 204.0 999.9 899972
374320 190157 176742 92.9 32467657.9 170.7 999.9 529674
431960 190129 186792 98.2 19547974.7 102.8 999.8 802753
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$

```

Figure 8: The ten games with the most reviews. Columns: app_id, reviews, recommended, positive %, total hours, average hours, maximum hours, helpful votes.

app_id	Game	Reviews	Positive %	Avg. hours	Helpful votes
440	Team Fortress 2	319,492	92.3	318.7	521,978
252490	Rust	270,684	83.6	343.4	771,227
1091500	Cyberpunk 2077	226,414	74.5	102.2	1,189,869
730	Counter-Strike: Global Offensive	219,737	84.8	429.0	1,910,071
570	Dota 2	216,914	79.2	429.1	842,346
444090	Paladins®	204,176	84.0	139.8	188,983
292030	The Witcher® 3: Wild Hunt	204,166	94.6	154.4	813,793
377160	Fallout 4	196,789	81.6	204.0	899,972
374320	DARK SOULS™ III	190,157	92.9	170.7	529,674
431960	Wallpaper Engine	190,129	98.2	102.8	802,753

Table 3: The same ten games with their titles looked up from games.csv.

The maximum-hours column reads 999.9 for almost every popular game. That is not a bug in the job: the hours field in the source data never goes above 1,000, so for any game with a large player base at least one reviewer sits at the cap.

5.2 Binning Games by Rating

Performs binning on the Steam rating label, writing every game into one output file per bin. The program is BinningByRating.

This is the **binning** pattern from the data organization family. The job is map-only: each mapper decides which bin a record belongs to and writes it straight to that bin’s file through MultipleOutputs, so there is no shuffle, no sort and no reducer. The bin name comes from the rating column with the spaces removed (“Very Positive” becomes VeryPositive-m-00000), and a counter per bin records how many games landed in it. LazyOutputFormat stops Hadoop from also writing an empty default part-m-000000 file.

This analysis consists of:

1. **Driver Class** — sets zero reducers and registers the bins named output.
2. **Mapper Class** — uses MultipleOutputs to write each game to the file of its rating bin, and counts the games per bin.

Driver Class

```

1 package steam.binning;
2
3 import org.apache.hadoop.conf.Configuration;
4 import org.apache.hadoop.fs.Path;
5 import org.apache.hadoop.io.NullWritable;
6 import org.apache.hadoop.io.Text;
7 import org.apache.hadoop.mapreduce.Job;
8 import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
9 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
10 import org.apache.hadoop.mapreduce.lib.output.LazyOutputFormat;
11 import org.apache.hadoop.mapreduce.lib.output.MultipleOutputs;
12 import org.apache.hadoop.mapreduce.lib.output.TextOutputFormat;
13
14 /**
15  * MR-2 Binning games by rating (data organization pattern: binning).
16  *
17  * Map-only job. Every game is written to one of nine output files, one per
18  * Steam rating label, using MultipleOutputs. No reducer and no shuffle.
19  *
20  * Usage: hadoop jar BinningByRating.jar steam.binning.BinningByRating <in> <out>
21  */
22 public class BinningByRating {
23
24     public static void main(String[] args) throws Exception {
25         if (args.length != 2) {
26             System.err.println("Usage: BinningByRating <input> <output>");
27             System.exit(2);
28         }
29         Configuration conf = new Configuration();
30         Job job = Job.getInstance(conf, "Steam binning games by rating");
31         job.setJarByClass(BinningByRating.class);
32
33         job.setMapperClass(BinningMapper.class);
34         job.setNumReduceTasks(0);
35
36         job.setOutputKeyClass(Text.class);
37         job.setOutputValueClass(NullWritable.class);
38
39         MultipleOutputs.addNamedOutput(job, "bins", TextOutputFormat.class, Text.class, NullWritable.class);
40         // only create the bin files; skip the empty default part-m-00000
41         LazyOutputFormat.setOutputFormatClass(job, TextOutputFormat.class);
42
43         FileInputFormat.addInputPath(job, new Path(args[0]));
44         FileOutputFormat.setOutputPath(job, new Path(args[1]));
45         System.exit(job.waitForCompletion(true) ? 0 : 1);
46     }
47 }

```

Mapper Class

```

1 package steam.binning;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.io.LongWritable;
6 import org.apache.hadoop.io.NullWritable;
7 import org.apache.hadoop.io.Text;
8 import org.apache.hadoop.mapreduce.Mapper;
9 import org.apache.hadoop.mapreduce.lib.output.MultipleOutputs;
10
11 /**
12  * Reads games.csv and writes each game into the bin (output file) named after

```

```

13  * its Steam rating label, e.g. "Very Positive" -> VeryPositive-m-00000.
14  * A counter per bin records how many games landed in it.
15  */
16  public class BinningMapper extends Mapper<LongWritable, Text, Text, NullWritable> {
17
18      private MultipleOutputs<Text, NullWritable> mos;
19
20      @Override
21      protected void setup(Context context) {
22          mos = new MultipleOutputs<>(context);
23      }
24
25      @Override
26      protected void map(LongWritable key, Text value, Context context)
27          throws IOException, InterruptedException {
28          // app_id,title,date_release,win,mac,linux,rating,positive_ratio,user_reviews,
29          // price_final,price_original,discount,steam_deck
30          String[] f = value.toString().split(",");
31          if (f.length != 13 || f[0].equals("app_id")) {
32              return; // header or malformed row
33          }
34          String rating = f[6].trim();
35          String bin = rating.replaceAll("[^A-Za-z]", ""); // file names must be alphanumeric
36          mos.write("bins", value, NullWritable.get(), bin);
37          context.getCounter("Bins", rating).increment(1);
38      }
39
40      @Override
41      protected void cleanup(Context context) throws IOException, InterruptedException {
42          mos.close();
43      }
44  }

```

Execution

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ hadoop jar BinningByRating/BinningByRating.jar steam.binning.BinningByRating /steam/ga
mes.csv /steam/out_binning
2026-09-11 20:18:19,061 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
2026-09-11 20:18:19,504 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:18:19,968 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool inte
rface and execute your application with ToolRunner to remedy this.
2026-09-11 20:18:19,991 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/rii/.staging/
job_1789136053647_0001
2026-09-11 20:18:20,216 INFO input.FileInputFormat: Total input files to process : 1
2026-09-11 20:18:20,298 INFO mapreduce.JobSubmitter: number of splits:1
2026-09-11 20:18:20,457 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789136053647_0001
2026-09-11 20:18:20,457 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-11 20:18:20,612 INFO conf.Configuration: resource-types.xml not found
2026-09-11 20:18:20,612 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-11 20:18:21,412 INFO impl.YarnClientImpl: Submitted application application_1789136053647_0001
2026-09-11 20:18:21,444 INFO mapreduce.Job: The url to track the job: http://pop-os:8088/proxy/application_1789136053647_0001/
2026-09-11 20:18:21,444 INFO mapreduce.Job: Running job: job_1789136053647_0001
2026-09-11 20:18:29,550 INFO mapreduce.Job: Job job_1789136053647_0001 running in uber mode : false
2026-09-11 20:18:29,551 INFO mapreduce.Job: map 0% reduce 0%
2026-09-11 20:18:34,624 INFO mapreduce.Job: map 100% reduce 0%
2026-09-11 20:18:35,641 INFO mapreduce.Job: Job job_1789136053647_0001 completed successfully
[... 49 more lines ...]

```

Figure 9: Binning job running on games.csv.

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ hadoop jar BinningByRating/BinningByRating.jar steam.binning.BinningByRating /steam/ga
mes.csv /steam/out_binning
[... 16 lines above ...]
2026-09-11 20:18:35,641 INFO mapreduce.Job: Job job_1789136053647_0001 completed successfully
2026-09-11 20:18:35,704 INFO mapreduce.Job: Counters: 42
  File System Counters
    FILE: Number of bytes read=0
    FILE: Number of bytes written=310351
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=4863892
    HDFS: Number of bytes written=4863668
    HDFS: Number of read operations=15
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=18
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Launched map tasks=1
    Data-local map tasks=1
    Total time spent by all maps in occupied slots (ms)=8367
    Total time spent by all reduces in occupied slots (ms)=0
    Total time spent by all map tasks (ms)=2789
    Total vcore-milliseconds taken by all map tasks=2789
    Total megabyte-milliseconds taken by all map tasks=2141952
  Map-Reduce Framework
    Map input records=50873
    Map output records=0
    Input split bytes=102
    Spilled Records=0
    Failed Shuffles=0
    Merged Map outputs=0
    GC time elapsed (ms)=9
    CPU time spent (ms)=2680
    Physical memory (bytes) snapshot=316841984
    Virtual memory (bytes) snapshot=2559520768
    Total committed heap usage (bytes)=260046848
    Peak Map Physical memory (bytes)=316841984
    Peak Map Virtual memory (bytes)=2559520768
  Bins
    Mixed=12157
    Mostly Negative=1849
    Mostly Positive=8738
    Negative=303
    Overwhelmingly Negative=14
    Overwhelmingly Positive=1110
    Positive=13502
    Very Negative=60
    Very Positive=13139
  File Input Format Counters
    Bytes Read=4863790
  File Output Format Counters
    Bytes Written=0
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ █

```

Figure 10: Job counters, including the nine Bins counters.

One map task read the 50,873 lines of `games.csv` (50,872 games and the header). The Map output records counter is 0 because `MultipleOutputs` writes to the bin files directly instead of going through the normal map output. The Bins counters are the real result: the nine of them add up to 50,872, so every game was placed in exactly one bin.

Output

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ hdfs dfs -ls /steam/out_binning 56 hdfs dfs -cat /steam/out_binning/OverwhelminglyNeg
ative-m-00000
2026-09-11 20:18:45,564 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
Found 10 items
-rw-r--r-- 1 rii supergroup 1090434 2026-09-11 20:18 /steam/out_binning/Mixed-m-00000
-rw-r--r-- 1 rii supergroup 185177 2026-09-11 20:18 /steam/out_binning/MostlyNegative-m-00000
-rw-r--r-- 1 rii supergroup 872438 2026-09-11 20:18 /steam/out_binning/MostlyPositive-m-00000
-rw-r--r-- 1 rii supergroup 28067 2026-09-11 20:18 /steam/out_binning/Negative-m-00000
-rw-r--r-- 1 rii supergroup 1570 2026-09-11 20:18 /steam/out_binning/OverwhelminglyNegative-m-00000
-rw-r--r-- 1 rii supergroup 119286 2026-09-11 20:18 /steam/out_binning/OverwhelminglyPositive-m-00000
-rw-r--r-- 1 rii supergroup 1273597 2026-09-11 20:18 /steam/out_binning/Positive-m-00000
-rw-r--r-- 1 rii supergroup 6409 2026-09-11 20:18 /steam/out_binning/VeryNegative-m-00000
-rw-r--r-- 1 rii supergroup 1286690 2026-09-11 20:18 /steam/out_binning/VeryPositive-m-00000
-rw-r--r-- 1 rii supergroup 0 2026-09-11 20:18 /steam/out_binning/_SUCCESS
2026-09-11 20:18:47,024 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
1180320,War of the Three Kingdoms,2021-12-17,true,false,false,Overwhelmingly Negative,15,21276,0.0,0.0,0.0,true
1416420,Expansion - Europa Universalis IV: Leviathan,2021-04-27,true,true,true,Overwhelmingly Negative,10,4723,19.99,19.99,0.0,true
512032,Sid Meier's Civilization® VI: Vikings Scenario Pack,2016-12-20,true,true,true,Overwhelmingly Negative,15,1184,0.0,0.0,0.0,true
2357570,Overwatch® 2,2023-08-10,true,false,false,Overwhelmingly Negative,9,181198,0.0,0.0,0.0,true
1850380,Tricolour Lovestory TrueEnd,2022-01-27,true,false,false,Overwhelmingly Negative,14,1739,0.99,0.99,0.0,true
2016940,02Jam Online,2023-01-16,true,false,false,Overwhelmingly Negative,17,1689,0.0,0.0,0.0,true
1799220,Malice,2022-11-02,true,false,false,Overwhelmingly Negative,16,723,7.99,7.99,0.0,true
267600,Airport Simulator 2014,2013-12-19,true,false,false,Overwhelmingly Negative,13,553,4.99,4.99,0.0,true
1563130,SuperPower 3,2022-10-07,true,false,false,Overwhelmingly Negative,10,880,29.99,29.99,0.0,true
201510,Flatout 3: Chaos & Destruction,2011-12-13,true,false,false,Overwhelmingly Negative,15,2766,9.99,9.99,0.0,true
1162602,TEKKEN 7 - DLC13: Frame Data Display,2019-12-10,true,false,false,Overwhelmingly Negative,14,605,1.49,2.99,50.0,true
246090,Spacebase DF-9,2014-10-27,true,true,true,Overwhelmingly Negative,18,3261,9.99,9.99,0.0,true
573831,Tom Clancy's Ghost Recon® Wildlands - Narco Road,2017-04-25,true,false,false,Overwhelmingly Negative,16,564,14.99,14.99,0.0,tru
e
227160,Kinetic Void,2014-11-21,true,true,true,Overwhelmingly Negative,17,1381,19.99,19.99,0.0,true
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ █

```

Figure 11: The nine bin files in HDFS, and the smallest bin (Overwhelmingly Negative, 14 games) printed in full.

Bin (rating)	Games	Share
Positive	13,502	26.5%
Very Positive	13,139	25.8%
Mixed	12,157	23.9%
Mostly Positive	8,738	17.2%
Mostly Negative	1,849	3.6%
Overwhelmingly Positive	1,110	2.2%
Negative	303	0.6%
Very Negative	60	0.1%
Overwhelmingly Negative	14	0.03%
Total	50,872	100%

Table 4: Games per bin, from the job's Bins counters.

5.3 Games Joined with their Reviews

Returns each game's title and rating together with how many people reviewed it and how many hours they played it (GameReviewJoin).

This is a **reduce-side join**. The two inputs have different layouts, so each gets its own mapper through MultipleInputs. Both mappers use app_id as the key and tag their values — G for a game record, R for review data — so that the reducer, which receives everything for one app_id together, can tell the two sides apart and put them back together.

Sending all 41 million reviews through the shuffle one by one would be slow, so the review mapper uses **in-mapper combining**: it keeps a running total per game in a hash map and emits a single R record per game when its input split is done. The join is an inner join — games nobody reviewed, and reviews whose game is missing from games.csv, are dropped and counted.

This analysis consists of:

1. **Driver Class** — registers the two inputs, each with its own mapper, and one reducer.
2. **Game Mapper** — emits app_id and "G, title, rating".
3. **Recommendation Mapper** — adds up reviews, recommendations and hours per game in memory and emits app_id and "R, reviews, recommended, hours".
4. **Join Reducer** — matches the G record with the R records of the same game and emits the joined line.

Driver Class

```

1 package steam.join;
2
3 import org.apache.hadoop.conf.Configuration;
4 import org.apache.hadoop.fs.Path;
5 import org.apache.hadoop.io.Text;
6 import org.apache.hadoop.mapreduce.Job;
7 import org.apache.hadoop.mapreduce.lib.input.MultipleInputs;
8 import org.apache.hadoop.mapreduce.lib.input.TextInputFormat;
9 import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
10
11 /**
12  * MR-3 Games joined with their reviews (join pattern: reduce-side join).
13  *
14  * Two inputs, each with its own mapper (MultipleInputs), meet in the reducer
15  * on app_id. The result puts each game's title and rating next to how many
16  * people reviewed it and how many hours they played.
17  *
18  * Usage: hadoop jar GameReviewJoin.jar steam.join.GameReviewJoin <games> <recommendations> <out>
19  */
20 public class GameReviewJoin {
21
22     public static void main(String[] args) throws Exception {
23         if (args.length != 3) {
24             System.err.println("Usage: GameReviewJoin <games> <recommendations> <output>");
25             System.exit(2);
26         }
27         Configuration conf = new Configuration();
28         Job job = Job.getInstance(conf, "Steam games joined with reviews");
29         job.setJarByClass(GameReviewJoin.class);
30
31         MultipleInputs.addInputPath(job, new Path(args[0]), TextInputFormat.class, GameMapper.class);
32         MultipleInputs.addInputPath(job, new Path(args[1]), TextInputFormat.class, RecommendationMapper.class);
33         job.setReducerClass(JoinReducer.class);
34
35         job.setOutputKeyClass(Text.class);
36         job.setOutputValueClass(Text.class);
37
38         FileOutputFormat.setOutputPath(job, new Path(args[2]));
39         System.exit(job.waitForCompletion(true) ? 0 : 1);
40     }
41 }

```

Game Mapper

```

1 package steam.join;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.io.LongWritable;
6 import org.apache.hadoop.io.Text;
7 import org.apache.hadoop.mapreduce.Mapper;

```

```

8
9 /**
10  * Left side of the join: games.csv.
11  * Emits (app_id, "G<TAB>title<TAB>rating") so the reducer can tell a game
12  * record apart from review records arriving under the same key.
13  */
14 public class GameMapper extends Mapper<LongWritable, Text, Text, Text> {
15
16     private final Text appId = new Text();
17     private final Text tagged = new Text();
18
19     @Override
20     protected void map(LongWritable key, Text value, Context context)
21         throws IOException, InterruptedException {
22         String[] f = value.toString().split(",");
23         if (f.length != 13 || f[0].equals("app_id")) {
24             return; // header or malformed row
25         }
26         appId.set(f[0]);
27         tagged.set("G\t" + f[1] + "\t" + f[6]);
28         context.write(appId, tagged);
29     }
30 }

```

Recommendation Mapper

```

1 package steam.join;
2
3 import java.io.IOException;
4 import java.util.HashMap;
5 import java.util.Map;
6
7 import org.apache.hadoop.io.LongWritable;
8 import org.apache.hadoop.io.Text;
9 import org.apache.hadoop.mapreduce.Mapper;
10
11 /**
12  * Right side of the join: recommendations.csv (41 million rows).
13  *
14  * Emitting one record per review would push 41 million records through the
15  * shuffle. Instead the mapper keeps a running total per app_id in memory
16  * (in-mapper combining) and emits one "R<TAB>reviews<TAB>recommended<TAB>hours"
17  * record per game when its input split is finished.
18  */
19 public class RecommendationMapper extends Mapper<LongWritable, Text, Text, Text> {
20
21     // app_id -> {reviews, recommended, hours}
22     private final Map<String, double[]> totals = new HashMap<>();
23
24     @Override
25     protected void map(LongWritable key, Text value, Context context) {
26         // app_id,helpful,funny,date,is_recommended,hours,user_id,review_id
27         String[] f = value.toString().split(",");
28         if (f.length != 8 || f[0].equals("app_id")) {
29             return; // header or malformed row
30         }
31         try {
32             double hours = Double.parseDouble(f[5]);
33             double[] t = totals.computeIfAbsent(f[0], k -> new double[3]);
34             t[0] += 1;
35             t[1] += Boolean.parseBoolean(f[4]) ? 1 : 0;
36             t[2] += hours;
37         } catch (NumberFormatException e) {
38             context.getCounter("Join", "Malformed review rows").increment(1);
39         }
40     }
41 }

```

```

39     }
40 }
41
42 @Override
43 protected void cleanup(Context context) throws IOException, InterruptedException {
44     Text appId = new Text();
45     Text tagged = new Text();
46     for (Map.Entry<String, double[]> e : totals.entrySet()) {
47         double[] t = e.getValue();
48         appId.set(e.getKey());
49         tagged.set("R\t" + (long) t[0] + "\t" + (long) t[1] + "\t" + t[2]);
50         context.write(appId, tagged);
51     }
52 }
53 }

```

Join Reducer

```

1 package steam.join;
2
3 import java.io.IOException;
4
5 import org.apache.hadoop.io.Text;
6 import org.apache.hadoop.mapreduce.Reducer;
7
8 /**
9  * Inner join on app_id. For each game the reducer receives one "G" record
10  * (title, rating) and several "R" records (partial review totals from each
11  * mapper). It adds up the R records and writes one joined line:
12  *
13  * app_id title rating reviews positive% total_hours avg_hours
14  *
15  * Games with no reviews, and reviews whose game is missing from games.csv,
16  * are dropped and counted.
17  */
18 public class JoinReducer extends Reducer<Text, Text, Text, Text> {
19
20     private final Text out = new Text();
21
22     @Override
23     protected void reduce(Text key, Iterable<Text> values, Context context)
24         throws IOException, InterruptedException {
25         String title = null;
26         String rating = null;
27         long reviews = 0;
28         long recommended = 0;
29         double hours = 0;
30
31         for (Text v : values) {
32             String[] f = v.toString().split("\t");
33             if (f[0].equals("G")) {
34                 title = f[1];
35                 rating = f[2];
36             } else {
37                 reviews += Long.parseLong(f[1]);
38                 recommended += Long.parseLong(f[2]);
39                 hours += Double.parseDouble(f[3]);
40             }
41         }
42
43         if (title == null) {
44             context.getCounter("Join", "Reviews with no matching game").increment(reviews);
45             return;
46         }

```

```

47     if (reviews == 0) {
48         context.getCounter("Join", "Games with no reviews").increment(1);
49         return;
50     }
51     context.getCounter("Join", "Joined games").increment(1);
52     out.set(title + "\t" + rating + "\t" + reviews + "\t"
53         + String.format("%.1f", 100.0 * recommended / reviews) + "\t"
54         + String.format("%.1f", hours) + "\t"
55         + String.format("%.1f", hours / reviews));
56     context.write(key, out);
57 }
58 }

```

Execution

```

rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ time hadoop jar GameReviewJoin/GameReviewJoin.jar steam.join.GameReviewJoin /steam/games.csv /steam/recommendations.csv /steam/out_join
2026-09-11 20:21:27,256 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-11 20:21:27,723 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:21:28,126 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-11 20:21:28,151 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/rii/.staging/job_1789136053647_0003
2026-09-11 20:21:28,457 INFO input.FileInputFormat: Total input files to process : 1
2026-09-11 20:21:28,484 INFO input.FileInputFormat: Total input files to process : 1
2026-09-11 20:21:28,549 INFO mapreduce.JobSubmitter: number of splits:16
2026-09-11 20:21:29,174 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789136053647_0003
2026-09-11 20:21:29,174 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-11 20:21:29,346 INFO conf.Configuration: resource-types.xml not found
2026-09-11 20:21:29,347 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-11 20:21:29,467 INFO impl.YarnClientImpl: Submitted application application_1789136053647_0003
2026-09-11 20:21:29,500 INFO mapreduce.Job: The url to track the job: http://pop-os:8088/proxy/application_1789136053647_0003/
2026-09-11 20:21:29,501 INFO mapreduce.Job: Running job: job_1789136053647_0003
2026-09-11 20:21:34,597 INFO mapreduce.Job: Job job_1789136053647_0003 running in uber mode : false
2026-09-11 20:21:34,598 INFO mapreduce.Job: map 0% reduce 0%
2026-09-11 20:21:49,024 INFO mapreduce.Job: map 19% reduce 0%
2026-09-11 20:21:50,035 INFO mapreduce.Job: map 31% reduce 0%
2026-09-11 20:22:05,498 INFO mapreduce.Job: map 38% reduce 0%
2026-09-11 20:22:12,292 INFO mapreduce.Job: map 56% reduce 0%
2026-09-11 20:22:14,410 INFO mapreduce.Job: map 63% reduce 0%
2026-09-11 20:22:28,897 INFO mapreduce.Job: map 69% reduce 0%
2026-09-11 20:22:30,947 INFO mapreduce.Job: map 81% reduce 0%
2026-09-11 20:22:31,972 INFO mapreduce.Job: map 88% reduce 27%
2026-09-11 20:22:36,031 INFO mapreduce.Job: map 94% reduce 27%
2026-09-11 20:22:38,048 INFO mapreduce.Job: map 100% reduce 31%
2026-09-11 20:22:40,060 INFO mapreduce.Job: map 100% reduce 100%
2026-09-11 20:22:41,078 INFO mapreduce.Job: Job job_1789136053647_0003 completed successfully
[... 68 more lines ...]

```

Figure 12: Join job running over both inputs.

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ time hadoop jar GameReviewJoin/GameReviewJoin.jar steam.join.GameReviewJoin /steam/gam
es.csv /steam/recommendations.csv /steam/out_join
[... 27 lines above ...]
2026-09-11 20:22:41,078 INFO mapreduce.Job: Job job_1789136053647_0003 completed successfully
2026-09-11 20:22:41,734 INFO mapreduce.Job: Counters: 56
  File System Counters
    FILE: Number of bytes read=4928118
    FILE: Number of bytes written=15124883
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=2028148229
    HDFS: Number of bytes written=2155580
    HDFS: Number of read operations=53
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Launched map tasks=16
    Launched reduce tasks=1
    Data-local map tasks=16
    Total time spent by all maps in occupied slots (ms)=733170
    Total time spent by all reduces in occupied slots (ms)=114564
    Total time spent by all map tasks (ms)=244390
    Total time spent by all reduce tasks (ms)=28641
    Total vcore-milliseconds taken by all map tasks=244390
    Total vcore-milliseconds taken by all reduce tasks=28641
    Total megabyte-milliseconds taken by all map tasks=187691520
    Total megabyte-milliseconds taken by all reduce tasks=29328384
  Map-Reduce Framework
    Map input records=41205668
    Map output records=141638
    Map output bytes=4644815
    Map output materialized bytes=4928208
    Input split bytes=4028
    Combine input records=0
    Combine output records=0
    Reduce input groups=50872
    Reduce shuffle bytes=4928208
    Reduce input records=141638
    Reduce output records=37610
    Spilled Records=283276
    Shuffled Maps =16
    Failed Shuffles=0
    Merged Map outputs=16
    GC time elapsed (ms)=8002
    CPU time spent (ms)=147210
    Physical memory (bytes) snapshot=6722342912
    Virtual memory (bytes) snapshot=43278286848
    Total committed heap usage (bytes)=4762632192
    Peak Map Physical memory (bytes)=451522560
    Peak Map Virtual memory (bytes)=2541797376
    Peak Reduce Physical memory (bytes)=341549056
    Peak Reduce Virtual memory (bytes)=2787770368
  Join
    Games with no reviews=13262
    Joined games=37610
  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
  File Input Format Counters
    Bytes Read=0
  File Output Format Counters
    Bytes Written=2155580

real    1m15.075s
user    0m5.630s
sys     0m0.448s
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$

```

Figure 13: Job counters of the join, including the Join counters.

The job read 16 input splits — 15 from the review file and one from games.csv — and finished in 1 minute 15 seconds, faster than the summary job even though it reads the same 2 GB. The reason is in the counters: in-mapper combining cut the map output from 41 million records to 141,638, so the shuffle had almost nothing to move. The Join counters show 37,610 games matched and 13,262 games with no reviews at all; no review pointed to a game missing from games.csv.

Output

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ hdfs dfs -ls /steam/out_join && hdfs dfs -cat /steam/out_join/part-r-000000 | sort -t
$'\t' -k6 -gr | head -10
2026-09-11 20:23:22,534 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
Found 2 items
-rw-r--r-- 1 rii supergroup 0 2026-09-11 20:22 /steam/out_join/_SUCCESS
-rw-r--r-- 1 rii supergroup 2155580 2026-09-11 20:22 /steam/out_join/part-r-000000
2026-09-11 20:23:23,525 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
440 Team Fortress 2 Very Positive 319492 92.3 101809977.6 318.7
730 Counter-Strike: Global Offensive Very Positive 219737 84.8 94260307.3 429.0
570 Dota 2 Very Positive 216914 79.2 93072785.4 429.1
252490 Rust Very Positive 270684 83.6 92944004.6 343.4
359550 Tom Clancy's Rainbow Six® Siege Very Positive 189603 81.9 68445823.3 361.0
1085660 Destiny 2 Very Positive 185527 81.9 62504172.8 336.9
394360 Hearts of Iron IV Very Positive 143187 91.3 53086631.1 370.8
346110 ARK: Survival Evolved Very Positive 159736 79.6 46720359.4 292.5
230410 Warframe Very Positive 127812 88.0 46450175.3 363.4
271590 Grand Theft Auto V Very Positive 163909 68.3 46170394.5 281.7
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ █

```

Figure 14: The ten games with the most hours played. Columns: app_id, title, rating, reviews, positive %, total hours, average hours per reviewer.

Game	Rating	Reviews	Total hours	Avg. hours
Team Fortress 2	Very Positive	319,492	101,809,977.6	318.7
Counter-Strike: Global Offensive	Very Positive	219,737	94,260,307.3	429.0
Dota 2	Very Positive	216,914	93,072,785.4	429.1
Rust	Very Positive	270,684	92,944,004.6	343.4
Tom Clancy's Rainbow Six® Siege	Very Positive	189,603	68,445,823.3	361.0
Destiny 2	Very Positive	185,527	62,504,172.8	336.9
Hearts of Iron IV	Very Positive	143,187	53,086,631.1	370.8
ARK: Survival Evolved	Very Positive	159,736	46,720,359.4	292.5
Warframe	Very Positive	127,812	46,450,175.3	363.4
Grand Theft Auto V	Very Positive	163,909	46,170,394.5	281.7

Table 5: Top 10 games by total hours played, from the joined output.

6. Pig Analysis

All five scripts read the dataset straight from HDFS with `PigStorage(',')` and drop the header row with `FILTER ... BY app_id IS NOT NULL` — the header's `app_id` is the text “app_id”, which fails the cast to `int` and becomes null. Results are stored tab separated under `/steam/pig/`; a tab is used instead of a pipe because 104 game titles contain a `|` character.

Every ranking sorts *inside* a nested `FOREACH` block (`ORDER` then `LIMIT`) rather than with a top-level `ORDER BY`. For the “in each rating group” queries this is what the question needs — a separate top 10 per group — and for the overall rankings it avoids the sampling job that a top-level `ORDER BY` starts.

Each run ends with a Pig warning, `FIELD_DISCARDED_TYPE_CONVERSION_FAILED`. It counts the fields of the header rows that could not be cast to a number, and it is exactly how the header is filtered out, so it is expected rather than a data problem.

6.1 Top 5 Rating Groups

Returns the five Steam rating labels that the most games carry.

The games are grouped by rating and counted; the nine counts are then gathered into a single group (GROUP ... ALL) so they can be sorted together and cut to five. Pig compiles the script into two MapReduce jobs — the count, which runs with a combiner, and the ranking.

Script

```
1 -- Pig-1 Top 5 rating groups by number of games
2 games = LOAD '/steam/games.csv' USING PigStorage(',') AS
3     (app_id:int, title:chararray, date_release:chararray, win:chararray, mac:chararray,
4     linux:chararray, rating:chararray, positive_ratio:int, user_reviews:int,
5     price_final:double, price_original:double, discount:double, steam_deck:chararray);
6 valid_games = FILTER games BY app_id IS NOT NULL; -- drops the header row
7
8 grp_for_ratings = GROUP valid_games BY rating;
9 cnt_for_ratings = FOREACH grp_for_ratings GENERATE group AS rating, COUNT(valid_games) AS num_games;
10
11 grp_all = GROUP cnt_for_ratings ALL;
12 top5_for_ratings = FOREACH grp_all {
13     srt = ORDER cnt_for_ratings BY num_games DESC;
14     lim = LIMIT srt 5;
15     GENERATE FLATTEN(lim);
16 };
17 STORE top5_for_ratings INTO '/steam/pig/Top5Ratings' USING PigStorage('\t');
```

Execution

```

rrii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rrii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ pig -x mapreduce -f PigAnalysis/scripts/top5Ratings.pig
2026-09-11 20:23:41,559 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 20:23:41,560 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 20:23:41,560 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 20:23:41,591 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 20:23:41,591 [main] INFO org.apache.pig.Main - Logging error messages to: /home/rrii/Documents/Study/8th Semester/BDAL/Steam-Data-Analysis-using-Hadoop/pig_1789136621586.log
2026-09-11 20:23:41,695 [main] WARN org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-11 20:23:41,774 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file /home/rrii/.pigbootup not found
2026-09-11 20:23:41,815 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 20:23:41,815 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 20:23:42,143 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top5Ratings.pig-2f573a76-bc5d-4b83-a030-5991317e2334
2026-09-11 20:23:42,143 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 20:23:42,523 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-11 20:23:42,535 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-09-11 20:23:42,557 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 20:23:42,577 [main] INFO org.apache.pig.newplan.LogicalOptimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachCons
tantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 20:23:42,627 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (61 Old Gen) of size 1048576000 to monitor. collectionUsageThreshold = 734003200, us
ageThreshold = 734003200
2026-09-11 20:23:42,661 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 20:23:42,689 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 20:23:42,709 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 2
2026-09-11 20:23:42,709 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 2
2026-09-11 20:23:42,782 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:23:42,921 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-met
rics-publisher.enabled
2026-09-11 20:23:42,930 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 20:23:42,935 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markrese
t.buffer.percent
2026-09-11 20:23:42,935 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to defa
ult 0.3
2026-09-11 20:23:42,936 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 20:23:42,938 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-11 20:23:42,939 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionen
gine.mapReduceLayer.InputSizeReducerEstimator
2026-09-11 20:23:42,945 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSi
ze=4863790
2026-09-11 20:23:42,945 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-11 20:23:42,945 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2026-09-11 20:23:42,945 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-11 20:23:42,963 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-11 20:23:43,134 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/automaton-1.11-8.j
ar to DistributedCache through /tmp/temp-251705121/tmp-2822467333/pig-0.18.0-core-h3.jar
2026-09-11 20:23:43,173 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/automaton-1.11-8.j
ar to DistributedCache through /tmp/temp-251705121/tmp-629170201/automaton-1.11-8.jar
2026-09-11 20:23:43,202 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/antlr-runtime-3.4.
jar to DistributedCache through /tmp/temp-251705121/tmp-204110805/antlr-runtime-3.4.4.jar
2026-09-11 20:23:43,633 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/joda-time-2.14.0.j
ar to DistributedCache through /tmp/temp-251705121/tmp-1038689531/joda-time-2.14.0.jar
2026-09-11 20:23:44,072 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/RoaringBitmap-shad
ed-0.7.14.jar to DistributedCache through /tmp/temp-251705121/tmp407321936/RoaringBitmap-shaded-0.7.14.jar
2026-09-11 20:23:44,104 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-09-11 20:23:44,107 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-11 20:23:44,107 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-11 20:23:44,107 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-11 20:23:44,154 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-11 20:23:44,161 [JobControl] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:23:44,174 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-09-11 20:23:44,176 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.syst
em-metrics-publisher.enabled
2026-09-11 20:23:44,285 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/rrii/.staging/job_1789136053647_0
804
2026-09-11 20:23:44,302 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or JobSetJar(String).
2026-09-11 20:23:44,335 [JobControl] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2026-09-11 20:23:44,341 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 20:23:44,341 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-11 20:23:44,372 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-11 20:23:44,827 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-11 20:23:44,879 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.syst
em-metrics-publisher.enabled
2026-09-11 20:23:45,403 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1789136053647_0804
2026-09-11 20:23:45,403 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-11 20:23:45,502 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-11 20:23:45,549 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-11 20:23:45,550 [JobControl] INFO org.apache.hadoop.yarn.util.resource.ResourceUtils - Unable to find 'resource-types.xml'.
2026-09-11 20:23:45,830 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1789136053647_0804
2026-09-11 20:23:45,861 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://pop-os:8088/proxy/application_1789136053647_0804/
2026-09-11 20:23:45,862 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1789136053647_0804
2026-09-11 20:23:45,862 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases cnt_for_ratings,games,grp_for_ratings,valid_game
s
2026-09-11 20:23:45,862 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: games[2,8],games[-1,-1],valid_games[6,14],cn
t_for_ratings[9,18],grp_for_ratings[8,18] C: cnt_for_ratings[9,18],grp_for_ratings[8,18] R: cnt_for_ratings[9,18]
2026-09-11 20:23:45,877 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
[... 97 more lines ...]

```

Figure 15: top5Ratings.pig starting in mapreduce mode.

```

rrii@pop-os: ~/Steam-Data-Analysis-using-Hadoop$ pig -x mapreduce -f PigAnalysis/scripts/top5Ratings.pig
[... 115 lines above ...]
2026-09-11 20:24:35,421 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:24:35,436 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-09-11 20:24:35,449 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

HadoopVersion  PigVersion  UserId  StartedAt  FinishedAt  Features
3.4.1  0.18.0  rii  2026-09-11 20:23:42  2026-09-11 20:24:35  GROUP_BY,FILTER

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature  Outputs
job_1789136053647_0004  1  1  3  3  3  3  1  1  1  1  cnt_for_ratings,games,grp_for_ratings,valid_games  GROUP_BY,COMBINER
job_1789136053647_0005  1  1  5  5  5  5  4  4  4  4  grp_all,srt,top5_for_ratings  GROUP_BY  /steam/pig/Top5Ratings,

Input(s):
Successfully read 50873 records (4864154 bytes) from: "/steam/games.csv"

Output(s):
Successfully stored 5 records (89 bytes) in: "/steam/pig/Top5Ratings"

Counters:
Total records written : 5
Total bytes written : 89
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789136053647_0004 -> job_1789136053647_0005,
job_1789136053647_0005

2026-09-11 20:24:35,451 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:24:35,455 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:24:35,471 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:24:35,474 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:24:35,493 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:24:35,496 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:24:35,514 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:24:35,518 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:24:35,534 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:24:35,537 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:24:35,552 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:24:35,558 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:24:35,576 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 6 time
(s).
2026-09-11 20:24:35,576 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-11 20:24:35,593 [main] INFO org.apache.pig.Main - Pig script completed in 54 seconds and 81 milliseconds (54081 ms)
rrii@pop-os: ~/Steam-Data-Analysis-using-Hadoop$

```

Figure 16: End of the run: Success!, the two jobs, 50,873 records read and 5 stored.

Output

```

rrii@pop-os: ~/Steam-Data-Analysis-using-Hadoop$ hdfs dfs -cat /steam/pig/Top5Ratings/part-*
2026-09-11 20:24:36,177 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
Positive      13502
Very Positive 13139
Mixed         12157
Mostly Positive 8738
Mostly Negative 1849
rrii@pop-os: ~/Steam-Data-Analysis-using-Hadoop$

```

Figure 17: Top 5 rating groups.

The five counts are identical to the first five bins of the MapReduce binning job (Section 5.2).

Visualization

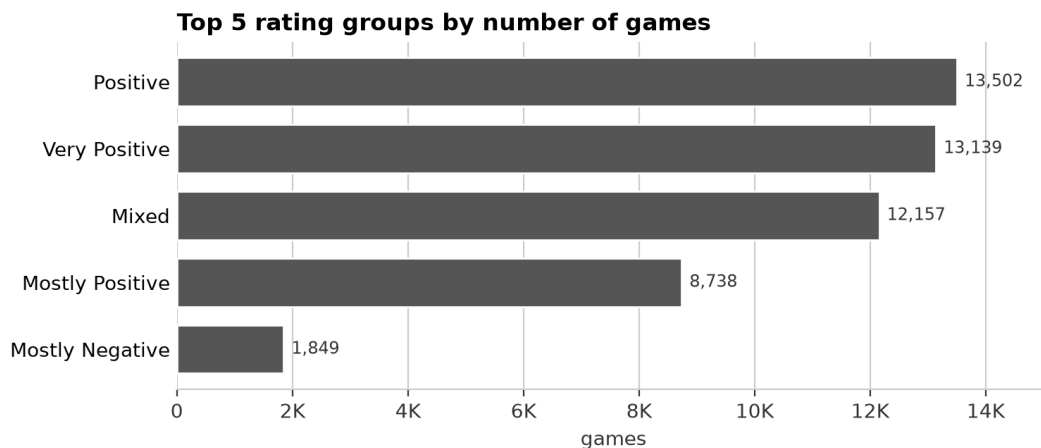


Figure 18: Top 5 rating groups by number of games.

6.2 Top 10 Most Reviewed Games

Returns the ten games with the most reviews on Steam (the user_reviews column).

Script

```

1  -- Pig-2 Top 10 most reviewed games
2  games = LOAD '/steam/games.csv' USING PigStorage(',') AS
3      (app_id:int, title:chararray, date_release:chararray, win:chararray, mac:chararray,
4      linux:chararray, rating:chararray, positive_ratio:int, user_reviews:int,
5      price_final:double, price_original:double, discount:double, steam_deck:chararray);
6  valid_games = FILTER games BY app_id IS NOT NULL; -- drops the header row
7
8  grp_all = GROUP valid_games ALL;
9  top10 = FOREACH grp_all {
10     srt = ORDER valid_games BY user_reviews DESC;
11     lim = LIMIT srt 10;
12     GENERATE FLATTEN(lim);
13 };
14 final_top10_reviewed = FOREACH top10 GENERATE app_id, title, rating, positive_ratio, user_reviews;
15 STORE final_top10_reviewed INTO '/steam/pig/Top10Reviewed' USING PigStorage('\t');

```

Execution

```

rrii@pop-os: ~/Steam-Data-Analysis-using-Hadoop

rrii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ pig -x mapreduce -f PigAnalysis/scripts/top10Reviewed.pig
2026-09-11 20:24:37,945 INFO org.apache.pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 20:24:37,946 INFO org.apache.pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 20:24:37,946 INFO org.apache.pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 20:24:37,984 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 20:24:37,984 [main] INFO org.apache.pig.Main - Logging error messages to: /home/rrii/Documents/Study/8th Semester/BDAL/Steam-Data-Analysis-using-Hadoop/pig_1789136677977.log
2026-09-11 20:24:38,190 [main] WARN org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-11 20:24:38,415 [main] INFO org.apache.pig.impl.util.Util - Default bootstrap file /home/rrii/.pigbootstrap not found
2026-09-11 20:24:38,585 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 20:24:38,585 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 20:24:39,124 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10Reviewed.pig-5e3cbb26-e2ec-4f9d-b5e0-f1332ec13174
2026-09-11 20:24:39,124 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 20:24:39,640 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-11 20:24:39,655 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY_FILTER
2026-09-11 20:24:39,676 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 20:24:39,788 [main] INFO org.apache.pig.newplan.LogicalOptimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachCons
tantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 20:24:39,754 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (61 Old Gen) of size 1048576000 to monitor. collectionUsageThreshold = 734003200, us
ageThreshold = 734003200
2026-09-11 20:24:39,790 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 20:24:39,811 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 20:24:39,811 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 20:24:39,908 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:24:40,067 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-met
rics-publisher.enabled
2026-09-11 20:24:40,079 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 20:24:40,084 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markrese
t.buffer.percent
2026-09-11 20:24:40,084 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to defa
ult 0.3
2026-09-11 20:24:40,086 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 20:24:40,088 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating % of required reducers.
2026-09-11 20:24:40,088 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-11 20:24:40,088 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2026-09-11 20:24:40,088 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-11 20:24:40,108 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-11 20:24:40,301 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/pig-0.18.0-core-h3.jar
to DistributedCache through /tmp/tmp-592995466/tmp602843273/pig-0.18.0-core-h3.jar
2026-09-11 20:24:40,739 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/automaton-1.11-8.j
ar to DistributedCache through /tmp/tmp-592995466/tmp274439624/automaton-1.11-8.jar
2026-09-11 20:24:40,768 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/antlr-runtime-3.4.
jar to DistributedCache through /tmp/tmp-592995466/tmp650850154/antlr-runtime-3.4.jar
2026-09-11 20:24:40,792 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/joda-time-2.14.0.j
ar to DistributedCache through /tmp/tmp-592995466/tmp108172220/joda-time-2.14.0.jar
2026-09-11 20:24:41,219 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/RoaringBitmap-shad
ed-0.7.14.jar to DistributedCache through /tmp/tmp-592995466/tmp-1793961007/RoaringBitmap-shaded-0.7.14.jar
2026-09-11 20:24:41,246 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-09-11 20:24:41,252 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-11 20:24:41,252 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-11 20:24:41,252 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-11 20:24:41,297 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-11 20:24:41,305 [JobControl] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:24:41,316 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-09-11 20:24:41,317 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.syst
em-metrics-publisher.enabled
2026-09-11 20:24:41,407 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/rrii/.staging/job_1789136053647_0
806
2026-09-11 20:24:41,417 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or JobSetJar(String).
2026-09-11 20:24:41,452 [JobControl] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2026-09-11 20:24:41,456 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 20:24:41,456 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-11 20:24:41,471 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-11 20:24:42,340 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-11 20:24:42,380 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.syst
em-metrics-publisher.enabled
2026-09-11 20:24:42,482 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1789136053647_0006
2026-09-11 20:24:42,483 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-11 20:24:42,583 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-11 20:24:42,636 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-11 20:24:42,637 [JobControl] INFO org.apache.hadoop.yarn.util.resource.ResourceUtils - Unable to find 'resource-types.xml'.
2026-09-11 20:24:42,699 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1789136053647_0006
2026-09-11 20:24:42,726 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://pop-os:8088/proxy/application_1789136053647_0006/
2026-09-11 20:24:42,727 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1789136053647_0006
2026-09-11 20:24:42,727 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases final_top10_reviewed, games, grp_all, srt, top10, val
id_games
2026-09-11 20:24:42,727 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: games[2,8],games[-1,-1],valid_games[6,14],gr
p_all[8,10] C: R: top10[9,8],srt[10,10],final_top10_reviewed[14,23]
2026-09-11 20:24:42,742 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
[... 48 more lines ...]

```

Figure 19: top10Reviewed.pig starting.

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ pig -x mapreduce -f PigAnalysis/scripts/top10Reviewed.pig
[... 71 lines above ...]
2026-09-11 20:25:25,080 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:25:25,292 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-09-11 20:25:25,310 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

HadoopVersion PigVersion UserId StartedAt FinishedAt Features
3.4.1 0.18.0 rii 2026-09-11 20:24:40 2026-09-11 20:25:25 GROUP_BY,FILTER

Success!

Job Stats (time in seconds):
JobId Maps Reduces MaxMapTime MinMapTime AvgMapTime MedianMapTime MaxReduceTime MinReduceTime AvgReduceTime MedianReduceTime Alias Feature Outputs
job_1789136053647_0006 1 1 3 3 3 3 15 15 15 15 final_top10_reviewed, games, grp_all, srt, top10, valid_games GROUP_BY /
steam/pig/Top10Reviewed,

Input(s):
Successfully read 50873 records (4864154 bytes) from: "/steam/games.csv"

Output(s):
Successfully stored 10 records (486 bytes) in: "/steam/pig/Top10Reviewed"

Counters:
Total records written : 10
Total bytes written : 486
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789136053647_0006

2026-09-11 20:25:25,331 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:25:25,340 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:25:25,417 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:25:25,430 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:25:25,504 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:25:25,516 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:25:25,573 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 6 time
(s).
2026-09-11 20:25:25,573 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-11 20:25:25,672 [main] INFO org.apache.pig.Main - Pig script completed in 47 seconds and 736 milliseconds (47736 ms)
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$

```

Figure 20: End of the run: one MapReduce job, 10 records stored in 48 seconds.

Output

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ hdfs dfs -cat /steam/pig/Top10Reviewed/part-*
2026-09-11 20:25:26,494 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
730 Counter-Strike: Global Offensive Very Positive 88 7494460
578080 PUBG: BATTLEGROUNDS Mixed 57 2217226
570 Dota 2 Very Positive 82 2045628
271590 Grand Theft Auto V Very Positive 86 1484122
359550 Tom Clancy's Rainbow Six® Siege Very Positive 86 993312
440 Team Fortress 2 Very Positive 93 985819
105600 Terraria Overwhelmingly Positive 97 943413
4000 Garry's Mod Overwhelmingly Positive 96 853733
252490 Rust Very Positive 87 786668
1172470 Apex Legends™ Very Positive 80 713182
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$

```

Figure 21: Top 10 most reviewed games. Columns: app_id, title, rating, positive ratio, user reviews.

Counter-Strike: Global Offensive has 7,494,460 reviews on Steam — more than three times the next game, PUBG: BATTLEGROUNDS (2,217,226). This column is Steam's own total for the whole store, which is why its order differs from MapReduce job 1: that job counts only the reviews present in recommendations.csv, where Team Fortress 2 comes first.

Visualization

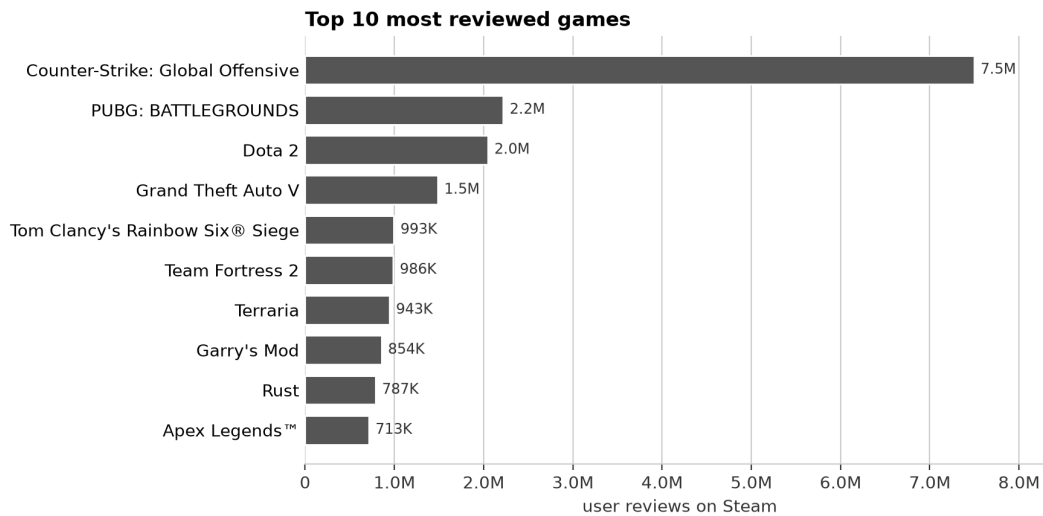


Figure 22: Top 10 most reviewed games.

6.3 Top 10 Most Reviewed Games in each Rating Group

Returns the ten most reviewed games inside each of the nine rating groups.

This is the nested-FOR EACH pattern: after GROUP BY rating, the block runs once per group, sorts that group's bag by user_reviews, keeps ten and FLATTENS them back into rows. The whole script runs as a single MapReduce job.

Script

```

1 -- Pig-3 Top 10 most reviewed games inside each rating group
2 games = LOAD '/steam/games.csv' USING PigStorage(',') AS
3   (app_id:int, title:chararray, date_release:chararray, win:chararray, mac:chararray,
4     linux:chararray, rating:chararray, positive_ratio:int, user_reviews:int,
5     price_final:double, price_original:double, discount:double, steam_deck:chararray);
6 valid_games = FILTER games BY app_id IS NOT NULL; -- drops the header row
7
8 grp_for_ratings = GROUP valid_games BY rating;
9 top10_reviewed_ratings = FOREACH grp_for_ratings {
10   sorted = ORDER valid_games BY user_reviews DESC;
11   top10 = LIMIT sorted 10;
12   GENERATE FLATTEN(top10);
13 };
14 top10_reviewed_by_rating = FOREACH top10_reviewed_ratings GENERATE rating, app_id, title, user_reviews;
15 STORE top10_reviewed_by_rating INTO '/steam/pig/Top10ReviewedByRating' USING PigStorage('\t');
```

Execution

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop

rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ pig -x mapreduce -f PigAnalysis/scripts/top10ReviewedByRating.pig
2026-09-11 20:25:29,770 INFO org.apache.pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 20:25:29,773 INFO org.apache.pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 20:25:29,773 INFO org.apache.pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 20:25:29,869 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 20:25:29,869 [main] INFO org.apache.pig.Main - Logging error messages to: /home/rii/Documents/Study/8th Semester/BDAL/Steam-Data-Analysis-using-Hadoop/pig_1789136729853.log
2026-09-11 20:25:30,067 [main] WARN org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-11 20:25:30,210 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file /home/rii/.pigbootstrap not found
2026-09-11 20:25:30,325 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 20:25:30,325 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 20:25:31,090 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10ReviewedByRating.pig-3dd4271-472f-4369-8c06-f150d686fb1b
2026-09-11 20:25:31,090 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 20:25:31,778 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-11 20:25:31,799 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,FILTER
2026-09-11 20:25:31,821 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 20:25:31,852 [main] INFO org.apache.pig.newplan.LogicalOptimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachCons
tantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 20:25:31,893 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (G1 Old Gen) of size 1048576000 to monitor. collectionUsageThreshold = 734003200, us
ageThreshold = 734003200
2026-09-11 20:25:31,939 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 20:25:31,963 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 20:25:31,963 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 20:25:32,084 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:25:32,354 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-met
rics-publisher.enabled
2026-09-11 20:25:32,376 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 20:25:32,387 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markrese
t.buffer.percent
2026-09-11 20:25:32,387 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to defa
ult 0.3
2026-09-11 20:25:32,391 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 20:25:32,395 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-11 20:25:32,396 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionen
gine.mapReduceLayer.InputSizeReducerEstimator
2026-09-11 20:25:32,405 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSi
ze=4863790
2026-09-11 20:25:32,406 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-11 20:25:32,406 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2026-09-11 20:25:32,406 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-11 20:25:32,423 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-11 20:25:32,743 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rii/bigdata/pig-0.18.0/pig-0.18.0-core-h3.jar
to DistributedCache through /tmp/tmp-518811992/tmp-274605462/pig-0.18.0-core-h3.jar
2026-09-11 20:25:32,772 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rii/bigdata/pig-0.18.0/lib/automaton-1.11-8.3
.jar to DistributedCache through /tmp/tmp-518811992/tmp117899704/automaton-1.11-8.3.jar
2026-09-11 20:25:32,801 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rii/bigdata/pig-0.18.0/lib/antlr-runtime-3.4.
.jar to DistributedCache through /tmp/tmp-518811992/tmp604300442/antlr-runtime-3.4.jar
2026-09-11 20:25:32,829 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rii/bigdata/pig-0.18.0/lib/joda-time-2.14.0.3
.jar to DistributedCache through /tmp/tmp-518811992/tmp1229277956/joda-time-2.14.0.jar
2026-09-11 20:25:32,857 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rii/bigdata/pig-0.18.0/lib/roaringbitmap-shad
ed-0.7.14.jar to DistributedCache through /tmp/tmp-518811992/tmp-135405322/RoaringBitmap-shaded-0.7.14.jar
2026-09-11 20:25:32,894 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-09-11 20:25:32,898 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-11 20:25:32,898 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-11 20:25:32,899 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-11 20:25:33,017 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-11 20:25:33,032 [JobControl] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:25:33,061 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-09-11 20:25:33,066 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.syst
em-metrics-publisher.enabled
2026-09-11 20:25:33,249 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/rii/.staging/job_1789136053647_0
007
2026-09-11 20:25:33,264 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or JobSetJar(String).
2026-09-11 20:25:33,347 [JobControl] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2026-09-11 20:25:33,359 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 20:25:33,360 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-11 20:25:33,420 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-11 20:25:33,896 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-11 20:25:33,940 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.syst
em-metrics-publisher.enabled
2026-09-11 20:25:34,547 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1789136053647_0007
2026-09-11 20:25:34,547 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-11 20:25:34,681 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-11 20:25:34,744 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-11 20:25:34,745 [JobControl] INFO org.apache.hadoop.yarn.util.resource.ResourceUtils - Unable to find 'resource-types.xml'.
2026-09-11 20:25:34,820 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1789136053647_0007
2026-09-11 20:25:34,860 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://pop-os:8088/proxy/application_1789136053647_0007/
2026-09-11 20:25:34,861 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1789136053647_0007
2026-09-11 20:25:34,861 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases games,grp_for_ratings,sorted,top10_reviewed_by_r
ating,top10_reviewed_ratings,valid_games
2026-09-11 20:25:34,861 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: games[2,0],games[-1,-1],valid_games[6,14],gr
p_for_ratings[8,10] C: R: top10_reviewed_ratings[9,25],sorted[10,13],top10_reviewed_by_rating[14,27]
2026-09-11 20:25:34,875 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
[... 48 more lines ...]

```

Figure 23: top10ReviewedByRating.pig starting.

```

rrii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rrii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ pig -x mapreduce -f PigAnalysis/scripts/top10ReviewedByRating.pig
[... 73 lines above ...]
2026-09-11 20:26:00,345 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:26:00,382 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-09-11 20:26:00,384 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

HadoopVersion  PigVersion  UserId  StartedAt  FinishedAt  Features
3.4.1  0.18.0  rii  2026-09-11 20:25:32  2026-09-11 20:26:00  GROUP_BY,FILTER

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
job_1789136053647_0007  1  1  4  4  4  3  3  3  3  games,grp_for_ratings,sorted,top10_reviewed_by_rating,top10_reviewed_ratings,valid
d_games GROUP_BY /steam/pig/Top10ReviewedByRating,

Input(s):
Successfully read 50873 records (4864154 bytes) from: "/steam/games.csv"

Output(s):
Successfully stored 90 records (4333 bytes) in: "/steam/pig/Top10ReviewedByRating"

Counters:
Total records written : 90
Total bytes written : 4333
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789136053647_0007

2026-09-11 20:26:00,387 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:26:00,391 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:26:00,407 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:26:00,413 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:26:00,436 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:26:00,444 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:26:00,464 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 6 time
(s).
2026-09-11 20:26:00,464 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-11 20:26:00,481 [main] INFO org.apache.pig.Main - Pig script completed in 30 seconds and 848 milliseconds (30848 ms)
rrii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ █

```

Figure 24: End of the run: 90 records stored — nine groups of ten.

Output

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ hdfs dfs -cat /steam/pig/Top10ReviewedByRating/part-1
2026-09-11 20:26:01,175 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
Mixed 578080 PUBG: BATTLEGROUNDS 2217226
Mixed 1938090 Call of Duty® 429206
Mixed 433850 Z1 Battle Royale 207328
Mixed 1468810 鬼谷八荒 Tale of Immortal 185051
Mixed 1517290 Battlefield™ 2042 153151
Mixed 227940 Heroes & Generals 141919
Mixed 1811260 EA SPORTS™ FIFA 23 113742
Mixed 304390 FOR HONOR™ 82277
Mixed 291480 Warface 80549
Mixed 397900 Business Tour - Board Game with Online Multiplayer 65424
Negative 364710 Wonky Pigeon! 47
Negative 273800 Forestry 2017 - The Simulation 47
Negative 345710 Damned Nation Reborn 45
Negative 1829586 HITMAN 3 - Seven Deadly Sins Act 7: Wrath 44
Negative 310880 Basketball Pro Management 2015 44
Negative 210490 Fray: Reloaded Edition 42
Negative 925670 Dungeon Kitty 42
Negative 280380 Uprising44: The Silent Shadows 42
Negative 519180 Streamer Simulator 42
Negative 315330 All Guns On Deck 42
Positive 591760 HyperBrawl Tournament 49
Positive 1710750 Slit Mouthed - 口裂け女 49
Positive 1947750 Queen's Wish 2: The Tormentor 49
Positive 1727930 Gorge 49
Positive 1260130 Banana Hell 49
Positive 1202770 Kirakira stars idol project AI 49
Positive 769680 姜神降临? 49
Positive 1714190 Breakout: Recharged 49
Positive 2167451 Monster Hunter Rise - 'Ran Page' Hunter layered armor set 49
Positive 588930 King 'n Knight 49
Very Negative 2338770 NBA 2K24 439
Very Negative 422420 Godus Wars 425
Very Negative 238870 Citadels 365
Very Negative 2280000 SQUARE ENIX AI Tech Preview: THE PORTOPIA SERIAL MURDER CASE 341
Very Negative 2205894 Wo Long: Fallen Dynasty Battle of Zhongyuan 338
Very Negative 246940 Lords of the Black Sun 324
Very Negative 273770 Game Tycoon 1.5 317
Very Negative 1090184 NTBS5 Top Secret Training Set - Season Pass 3 Characters 269
Very Negative 875950 CODE VEIN: Frozen Empress 267
Very Negative 252910 Skyscraper Simulator 262
Very Positive 730 Counter-Strike: Global Offensive 7494460
Very Positive 570 Dota 2 2045628
Very Positive 271590 Grand Theft Auto V 1484122
Very Positive 359550 Tom Clancy's Rainbow Six® Siege 993312
Very Positive 440 Team Fortress 2 985819
Very Positive 252490 Rust 786668
Very Positive 1172470 Apex Legends™ 713182
Very Positive 945360 Among Us 587821
Very Positive 1085660 Destiny 2 562723
Very Positive 1091500 Cyberpunk 2077 557051
Mostly Negative 1832640 Mirror 2: Project X 110981
Mostly Negative 1665460 eFootball™ 2024 59071
Mostly Negative 1272000 PAYDAY 3 29458
Mostly Negative 1962663 Call of Duty®: Warzone™ 2.0 19770
Mostly Negative 1367080 MOBILE SUIT GUNDAM BATTLE OPERATION 2 15685
Mostly Negative 1504570 Cultivation Tales 12299
Mostly Negative 730310 DYNASTY WARRIORS 9 9363
Mostly Negative 1945340 Destiny 2: Lightfall 7343
Mostly Negative 1336350 Elite Dangerous: Odyssey 7011
Mostly Negative 1816570 江湖十一 4772
Mostly Positive 236390 War Thunder 393681
Mostly Positive 221100 DayZ 296845
Mostly Positive 1063730 New World 222345
Mostly Positive 275850 No Man's Sky 209971
Mostly Positive 438100 VRChat 202265
Mostly Positive 1599340 Lost Ark 194889
Mostly Positive 1203220 NARAKA: BLADEPOINT 163924
Mostly Positive 1240440 Halo Infinite 161274
Mostly Positive 1238810 Battlefield™ V 141836
Mostly Positive 552990 World of Warships 129335
Overwhelmingly Negative 2357570 Overwatch® 2 181198
Overwhelmingly Negative 1180320 War of the Three Kingdoms 21276
Overwhelmingly Negative 1416420 Expansion - Europa Universalis IV: Leviathan 4723
Overwhelmingly Negative 246090 Spacebase DF-9 3261
Overwhelmingly Negative 201510 Flatout 3: Chaos & Destruction 2766
Overwhelmingly Negative 1850380 Tricolour Lovestory TrueEnd 1739
Overwhelmingly Negative 2016940 02Jam Online 1689
Overwhelmingly Negative 227160 Kinetic Void 1381
Overwhelmingly Negative 512032 Sid Meier's Civilization® VI: Vikings Scenario Pack 1184
Overwhelmingly Negative 1563130 SuperPower 3 880
Overwhelmingly Positive 105600 Terraria 943413
Overwhelmingly Positive 4000 Garry's Mod 853733
Overwhelmingly Positive 292030 The Witcher® 3: Wild Hunt 668455
Overwhelmingly Positive 431960 Wallpaper Engine 637341
Overwhelmingly Positive 550 Left 4 Dead 2 574470
Overwhelmingly Positive 413150 Stardew Valley 505802
Overwhelmingly Positive 227300 Euro Truck Simulator 2 494214
Overwhelmingly Positive 739630 Phasmophobia 486466
Overwhelmingly Positive 242760 The Forest 416113
Overwhelmingly Positive 892970 Valheim 356617
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ █

```

Figure 25: Top 10 most reviewed games in every rating group.

Two things stand out. The “Positive” and “Negative” groups never go above 49 reviews: Steam only gives those plain labels to games with 10–49 reviews, and a game that collects more moves up to “Very” or “Overwhelmingly”. And the most reviewed game in the Overwhelmingly Negative group is Overwatch® 2 with 181,198 reviews — a very large game can sit in the worst group.

Visualization

Top 10 most reviewed games in each rating group

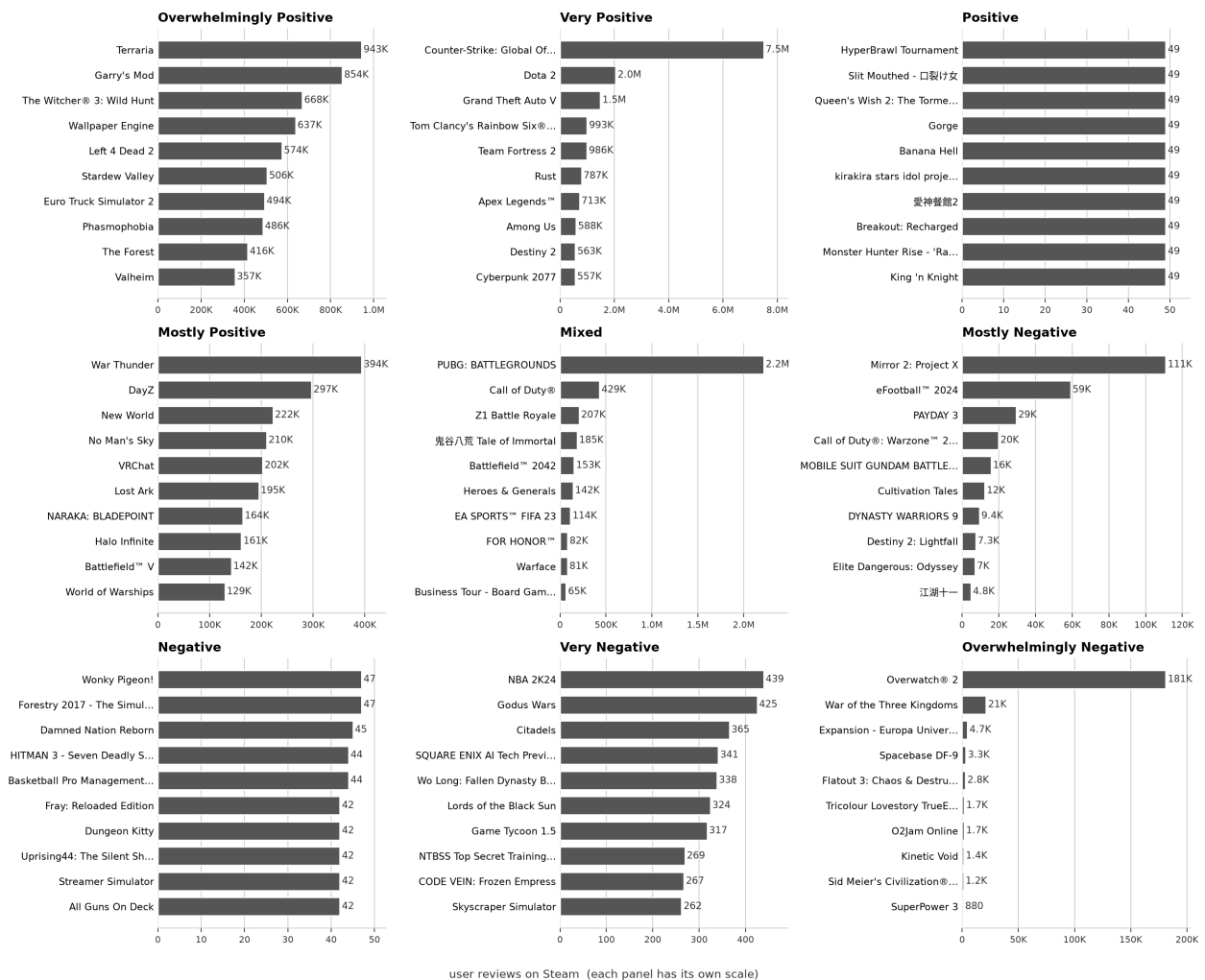


Figure 26: Top 10 most reviewed games in each rating group. Each panel has its own scale.

6.4 Top 10 Games by Total Hours Played

Returns the ten games whose reviewers played them for the most hours in total.

This script reads the full 2 GB review table. It adds up the hours per game with `GROUP BY app_id` and `SUM`, then brings in each game's title and rating with a **replicated join**: `games.csv` is small enough to be copied to every mapper, so the join happens on the map side without a second shuffle. The hours stay numeric for the sort and are only formatted with `SPRINTF` for the output.

Pig turns the script into a chain of three MapReduce jobs (visible in the Job Stats of the screenshot below): a map-only job that prepares `games.csv` for the replicated join, one job with 15 maps and 3 reducers that groups the 41 million reviews and performs the join, and a final job that ranks the result.

Script

```

1  -- Pig-4 Top 10 games by total hours played by their reviewers
2  recs = LOAD '/steam/recommendations.csv' USING PigStorage(',') AS
3      (app_id:int, helpful:int, funny:int, date:chararray, is_recommended:chararray,
4       hours:double, user_id:int, review_id:int);
5  valid_rec = FILTER recs BY app_id IS NOT NULL;    -- drops the header row
6
7  grp_for_games = GROUP valid_rec BY app_id;
8  played = FOREACH grp_for_games GENERATE group AS app_id,
9      COUNT(valid_rec) AS reviews,
10     SUM(valid_rec.hours) AS hours;
11
12 games = LOAD '/steam/games.csv' USING PigStorage(',') AS
13     (app_id:int, title:chararray, date_release:chararray, win:chararray, mac:chararray,
14     linux:chararray, rating:chararray, positive_ratio:int, user_reviews:int,
15     price_final:double, price_original:double, discount:double, steam_deck:chararray);
16 game_info = FOREACH (FILTER games BY app_id IS NOT NULL) GENERATE app_id, title, rating;
17
18 -- games.csv is small, so it is copied to every mapper (map-side replicated join)
19 joined = JOIN played BY app_id, game_info BY app_id USING 'replicated';
20 result = FOREACH joined GENERATE played::app_id AS app_id, game_info::title AS title,
21     game_info::rating AS rating, played::reviews AS reviews,
22     played::hours AS hours;
23
24 grp_all = GROUP result ALL;
25 top10 = FOREACH grp_all {
26     srt = ORDER result BY hours DESC;
27     lim = LIMIT srt 10;
28     GENERATE FLATTEN(lim);
29 };
30 -- hours stay numeric for the sort and are formatted only for output
31 final_top10_played = FOREACH top10 GENERATE app_id, title, rating, reviews,
32     SPRINTF('%.1f', hours) AS hours;
33 STORE final_top10_played INTO '/steam/pig/Top10Played' USING PigStorage('\t');

```

Execution

```

rrii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rrii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ pig -x mapreduce -f PigAnalysis/scripts/top10Played.pig
2026-09-11 20:38:16,921 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 20:38:16,922 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 20:38:16,922 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 20:38:16,952 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 20:38:16,952 [main] INFO org.apache.pig.Main - Logging error messages to: /home/rrii/Documents/Study/8th Semester/BDAL/Steam-Data-Analysis-using-Hadoop/pig_1789137496946.log
2026-09-11 20:38:17,071 [main] WARN org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-11 20:38:17,147 [main] INFO org.apache.pig.impl.util.Util - Default bootup file /home/rrii/.pigbootup not found
2026-09-11 20:38:17,194 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 20:38:17,195 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 20:38:17,515 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10Played.pig-74767eec-2546-46fb-9d7c-bb97fffa33b6
2026-09-11 20:38:17,515 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 20:38:17,960 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-11 20:38:17,976 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: REPLICATED_JOIN, GROUP_BY, FILTER
2026-09-11 20:38:17,993 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 20:38:18,010 [main] INFO org.apache.pig.newplan.Logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachCons
tantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 20:38:18,029 [main] INFO org.apache.pig.newplan.Logical.rules.ColumnPruneVisitor - Columns pruned for games: $2, $3, $4, $5, $7, $8, $9, $10, $11, $12
2026-09-11 20:38:18,063 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (G1 Old Gen) of size 1048576000 to monitor. collectionUsageThreshold = 734003200, us
ageThreshold = 734003200
2026-09-11 20:38:18,120 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 20:38:18,163 [main] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2026-09-11 20:38:18,174 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 20:38:18,177 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-11 20:38:18,225 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-11 20:38:18,225 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - number of input files: 1
2026-09-11 20:38:18,231 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombineOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 20:38:18,247 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 3
2026-09-11 20:38:18,247 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 3
2026-09-11 20:38:18,300 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMAFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:38:18,399 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-met
rics-publisher.enabled
2026-09-11 20:38:18,406 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScripState - Pig script settings are added to the job
2026-09-11 20:38:18,409 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markrese
t.buffer.percent
2026-09-11 20:38:18,409 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to defa
ult 0.3
2026-09-11 20:38:18,410 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 20:38:18,411 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-11 20:38:18,418 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-11 20:38:18,545 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/pig-0.18.0-core-h3.jar
to DistributedCache through /tmp/temp1054787997/tmp-1692456419/pig-0.18.0-core-h3.jar
2026-09-11 20:38:18,975 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/automaton-1.11-8.3
.jar to DistributedCache through /tmp/temp1054787997/tmp-1517125383/automaton-1.11-8.3.jar
2026-09-11 20:38:19,002 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/antlr-runtime-3.4.
.jar to DistributedCache through /tmp/temp1054787997/tmp-1923238425/antlr-runtime-3.4.jar
2026-09-11 20:38:19,431 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/joda-time-2.14.0.3
.jar to DistributedCache through /tmp/temp1054787997/tmp-858905376/joda-time-2.14.0.jar
2026-09-11 20:38:19,462 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/roaringbitmap-shad
ed-0.7.14.jar to DistributedCache through /tmp/temp1054787997/tmp1778862946/RoaringBitmap-shaded-0.7.14.jar
2026-09-11 20:38:19,492 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-09-11 20:38:19,495 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-11 20:38:19,495 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-11 20:38:19,495 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-11 20:38:19,523 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-11 20:38:19,529 [JobControl] INFO org.apache.hadoop.yarn.client.DefaultNoHARMAFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:38:19,538 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-09-11 20:38:19,540 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.syst
em-metrics-publisher.enabled
2026-09-11 20:38:19,647 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/rrii/.staging/job_1789136053647_0
014
2026-09-11 20:38:19,656 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or JobSetJar(String).
2026-09-11 20:38:19,689 [JobControl] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2026-09-11 20:38:19,698 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 20:38:19,698 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-11 20:38:19,700 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-11 20:38:20,150 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-11 20:38:20,188 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.syst
em-metrics-publisher.enabled
2026-09-11 20:38:20,289 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1789136053647_0014
2026-09-11 20:38:20,289 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-11 20:38:20,383 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-11 20:38:20,435 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-11 20:38:20,435 [JobControl] INFO org.apache.hadoop.yarn.util.resource.ResourceUtils - Unable to find 'resource-types.xml'.
2026-09-11 20:38:20,492 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1789136053647_0014
2026-09-11 20:38:20,520 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://pop-os:8088/proxy/application_1789136053647_0014/
2026-09-11 20:38:20,520 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1789136053647_0014
2026-09-11 20:38:20,520 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases 1-2, game_info, games
2026-09-11 20:38:20,520 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: games[12,8], games[-1,-1], 1-2[16,21], game_inf
o[16,12] C: R:
2026-09-11 20:38:20,536 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
[... 150 more lines ...]

```

Figure 27: top10Played.pig starting.

```

rrii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rrii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ pig -x mapreduce -f PigAnalysis/scripts/top10Played.pig
[... 168 lines above ...]
2026-09-11 20:42:50,361 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:42:50,439 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-09-11 20:42:50,504 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

HadoopVersion  PigVersion  UserId  StartedAt  FinishedAt  Features
3.4.1  0.18.0  rii  2026-09-11 20:38:18  2026-09-11 20:42:50  REPLICATED_JOIN, GROUP_BY, FILTER

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature  Outputs
job_1789136053647_0014  1  0  6  6  6  0  0  1-2, game_info, games  MAP_ONLY
job_1789136053647_0015  15  3  66  36  52  140  65  47  grp_for_games, joined, played, recs, result, valid_rec
COMBINER  REPLICATED_JOIN, GROUP_BY,
job_1789136053647_0016  1  1  3  3  3  3  3  3  final_top10_played, grp_all, srt, top10  GROUP_BY  /steam/pig/Top10Played,

Input(s):
Successfully read 50873 records (4864154 bytes) from: "/steam/games.csv"
Successfully read 41154795 records (2023286021 bytes) from: "/steam/recommendations.csv"

Output(s):
Successfully stored 10 records (555 bytes) in: "/steam/pig/Top10Played"

Counters:
Total records written : 10
Total bytes written : 555
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789136053647_0014 -> job_1789136053647_0015,
job_1789136053647_0015 -> job_1789136053647_0016,
job_1789136053647_0016

2026-09-11 20:42:50,531 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:42:50,539 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:42:50,624 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:42:50,637 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:42:50,714 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:42:50,726 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:42:50,812 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:42:50,825 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:42:50,934 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:42:50,946 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:42:51,054 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:42:51,444 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:42:51,531 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:42:51,537 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:42:51,608 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:42:51,616 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:42:51,706 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:42:51,716 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:42:51,782 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 7 time
(s).
2026-09-11 20:42:51,782 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-11 20:42:51,889 [main] INFO org.apache.pig.Main - Pig script completed in 4 minutes, 34 seconds and 925 milliseconds (274925 ms)
rrii@pop-os:~/Steam-Data-Analysis-using-Hadoop$

```

Figure 28: End of the run: 41,154,795 review lines and 50,873 game lines read, 10 records stored, in 4 minutes 35 seconds.

Output

```

rrii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rrii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ hdfs dfs -cat /steam/pig/Top10Played/part-*
2026-09-11 20:42:52,686 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
440  Team Fortress 2 Very Positive 319492 101809977.6
730  Counter-Strike: Global Offensive Very Positive 219737 94260307.3
570  Dota 2 Very Positive 216914 93072785.4
252490 Rust Very Positive 270684 92944004.6
359550 Tom Clancy's Rainbow Six® Siege Very Positive 189603 68445823.3
1085660 Destiny 2 Very Positive 185527 62504172.8
394360 Hearts of Iron IV Very Positive 143187 53086631.1
346110 ARK: Survival Evolved Very Positive 159736 46720359.4
230410 Warframe Very Positive 127812 46450175.3
271590 Grand Theft Auto V Very Positive 163909 46170394.5
rrii@pop-os:~/Steam-Data-Analysis-using-Hadoop$

```

Figure 29: Top 10 games by total hours played. Columns: app_id, title, rating, reviews, total hours.

The list is identical to the top ten of the MapReduce join (Section 5.3) — same games, same order, same review counts and the same hours to one decimal place. The two tools took very different routes to it: the hand-written join finished in 1 minute 15 seconds, the Pig script in 4 minutes 35 seconds, mostly because Pig runs three jobs one after another while the Java version does everything in one job and shrinks its shuffle with in-mapper combining.

Visualization

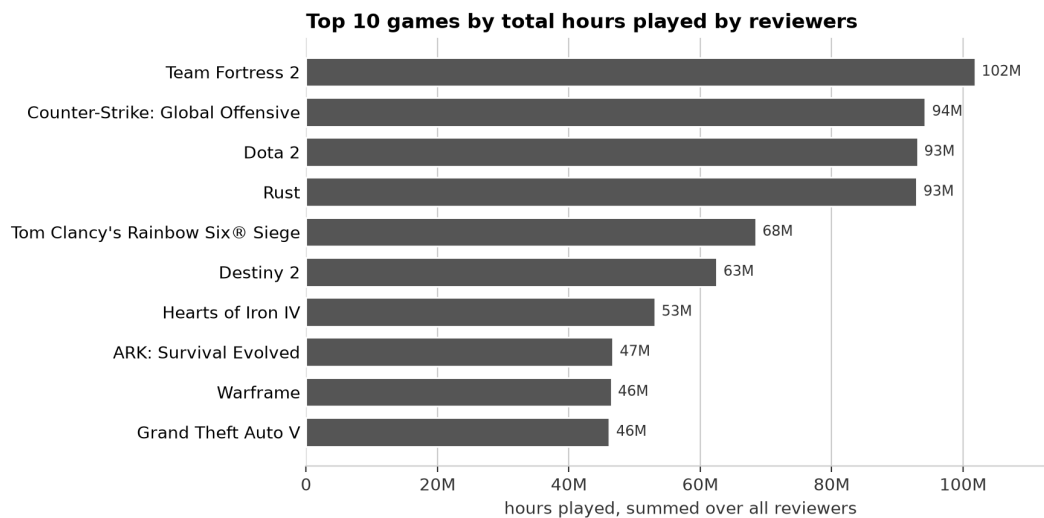


Figure 30: Top 10 games by total hours played by reviewers.

6.5 Top 10 Games by Total Hours Played in each Rating Group

Returns the ten most played games inside each rating group.

Same first half as the previous script — sum the hours per game and replicated-join the titles and ratings — followed by the nested sort of Section 6.3, grouped by rating.

Script

```

1  -- Pig-5 Top 10 games by total hours played inside each rating group
2  recs = LOAD '/steam/recommendations.csv' USING PigStorage(',') AS
3      (app_id:int, helpful:int, funny:int, date:chararray, is_recommended:chararray,
4      hours:double, user_id:int, review_id:int);
5  valid_rec = FILTER recs BY app_id IS NOT NULL;    -- drops the header row
6
7  grp_for_games = GROUP valid_rec BY app_id;
8  played = FOREACH grp_for_games GENERATE group AS app_id,
9      COUNT(valid_rec) AS reviews,
10     SUM(valid_rec.hours) AS hours;
11
12 games = LOAD '/steam/games.csv' USING PigStorage(',') AS
13     (app_id:int, title:chararray, date_release:chararray, win:chararray, mac:chararray,
14     linux:chararray, rating:chararray, positive_ratio:int, user_reviews:int,
15     price_final:double, price_original:double, discount:double, steam_deck:chararray);
16 game_info = FOREACH (FILTER games BY app_id IS NOT NULL) GENERATE app_id, title, rating;
17
18 joined = JOIN played BY app_id, game_info BY app_id USING 'replicated';
19 result = FOREACH joined GENERATE game_info::rating AS rating, played::app_id AS app_id,
20     game_info::title AS title, played::reviews AS reviews,
21     played::hours AS hours;
22
23 grp_for_ratings = GROUP result BY rating;
24 top10_played_ratings = FOREACH grp_for_ratings {
25     sorted = ORDER result BY hours DESC;
26     top10 = LIMIT sorted 10;
27     GENERATE FLATTEN(top10);
28 };
29 -- hours stay numeric for the sort and are formatted only for output
30 final_top10_played_by_rating = FOREACH top10_played_ratings GENERATE rating, app_id, title, reviews,
31     SPRINTF('%.1f', hours) AS hours;
32 STORE final_top10_played_by_rating INTO '/steam/pig/Top10PlayedByRating' USING PigStorage('\t');

```

Execution

```

rrii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rrii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ pig -x mapreduce -f PigAnalysis/scripts/top10PlayedByRating.pig
2026-09-11 20:42:55,115 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 20:42:55,117 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 20:42:55,117 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 20:42:55,234 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 20:42:55,234 [main] INFO org.apache.pig.Main - Logging error messages to: /home/rrii/Documents/Study/8th Semester/BDAL/Steam-Data-Analysis-using-Hadoop/pig_1789137775209.log
2026-09-11 20:42:55,438 [main] WARN org.apache.hadoop.util.NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-11 20:42:55,668 [main] INFO org.apache.pig.impl.util.Util - Default bootstrap file /home/rrii/.pigbootstrap not found
2026-09-11 20:42:55,784 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 20:42:55,784 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 20:42:56,415 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-top10PlayedByRating.pig-9fab957d-8e22-4aec-8533-2f92ac64a4b1
2026-09-11 20:42:56,416 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 20:42:57,073 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-11 20:42:57,087 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: REPLICATED_JOIN, GROUP_BY, FILTER
2026-09-11 20:42:57,108 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 20:42:57,137 [main] INFO org.apache.pig.newplan.Logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachCons
tantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer,
PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 20:42:57,170 [main] INFO org.apache.pig.newplan.Logical.rules.ColumnPruneVisitor - Columns pruned for games: $2, $3, $4, $5, $7, $8, $9, $10, $11, $12
2026-09-11 20:42:57,230 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (G1 Old Gen) of size 1048576000 to monitor. collectionUsageThreshold = 734003200, us
ageThreshold = 734003200
2026-09-11 20:42:57,295 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 20:42:57,331 [main] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2026-09-11 20:42:57,345 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 20:42:57,349 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-11 20:42:57,419 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-11 20:42:57,419 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - number of input files: 1
2026-09-11 20:42:57,429 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombineOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 20:42:57,453 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 3
2026-09-11 20:42:57,453 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 3
2026-09-11 20:42:57,539 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMAFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:42:57,674 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-met
rics-publisher.enabled
2026-09-11 20:42:57,680 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScripState - Pig script settings are added to the job
2026-09-11 20:42:57,684 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markrese
t.buffer.percent
2026-09-11 20:42:57,684 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to defa
ult 0.3
2026-09-11 20:42:57,685 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 20:42:57,686 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-11 20:42:57,693 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-11 20:42:57,861 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/pig-0.18.0-core-h3.jar
to DistributedCache through /tmp/tmp-297458409/tmp-907932092/pig-0.18.0-core-h3.jar
2026-09-11 20:42:58,293 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/automaton-1.11-8.3
.jar to DistributedCache through /tmp/tmp-297458409/tmp-205944096/automaton-1.11-8.3.jar
2026-09-11 20:42:58,322 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/antlr-runtime-3.4.
.jar to DistributedCache through /tmp/tmp-297458409/tmp-185356664/antlr-runtime-3.4.jar
2026-09-11 20:42:58,748 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/joda-time-2.14.0.3
.jar to DistributedCache through /tmp/tmp-297458409/tmp950172291/joda-time-2.14.0.jar
2026-09-11 20:42:59,180 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/home/rrii/bigdata/pig-0.18.0/lib/RoaringBitmap-shad
ed-0.7.14.jar to DistributedCache through /tmp/tmp-297458409/tmp60301136/RoaringBitmap-shaded-0.7.14.jar
2026-09-11 20:42:59,213 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-09-11 20:42:59,216 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-11 20:42:59,216 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-11 20:42:59,216 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-11 20:42:59,241 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-11 20:42:59,247 [JobControl] INFO org.apache.hadoop.yarn.client.DefaultNoHARMAFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:42:59,256 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-09-11 20:42:59,260 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.syst
em-metrics-publisher.enabled
2026-09-11 20:42:59,395 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/rrii/.staging/job_1789136053647_0
017
2026-09-11 20:42:59,421 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or JobSetJar(String).
2026-09-11 20:42:59,470 [JobControl] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2026-09-11 20:42:59,473 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 20:42:59,473 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-11 20:42:59,475 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2026-09-11 20:43:00,330 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-11 20:43:00,377 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.syst
em-metrics-publisher.enabled
2026-09-11 20:43:00,483 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1789136053647_0017
2026-09-11 20:43:00,483 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-11 20:43:00,578 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-11 20:43:00,627 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-11 20:43:00,627 [JobControl] INFO org.apache.hadoop.yarn.util.resource.ResourceUtils - Unable to find 'resource-types.xml'.
2026-09-11 20:43:00,680 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1789136053647_0017
2026-09-11 20:43:00,706 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://pop-os:8088/proxy/application_1789136053647_0017/
2026-09-11 20:43:00,707 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1789136053647_0017
2026-09-11 20:43:00,707 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases 1-2, game_info, games
2026-09-11 20:43:00,707 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: games[12,8], games[-1,-1], 1-2[16,21], game_inf
o[16,12] C: R:
2026-09-11 20:43:00,720 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
[... 150 more lines ...]

```

Figure 31: top10PlayedByRating.pig starting.

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ pig -x mapreduce -f PigAnalysis/scripts/top10PlayedByRating.pig
[... 168 lines above ...]
2026-09-11 20:46:57,275 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:46:57,352 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-09-11 20:46:57,417 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

HadoopVersion  PigVersion  UserId  StartedAt  FinishedAt  Features
3.4.1  0.18.0  rii  2026-09-11 20:42:57  2026-09-11 20:46:57  REPLICATED_JOIN, GROUP_BY, FILTER

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
job_1789136053647_0017  1  0  2  2  2  0  0  1-2,game_info,games  MAP_ONLY
job_1789136053647_0018  15  3  58  31  47  80  9  53  71  grp_for_games,joined,played,recs,result,valid_rec  REPLICATED_JOIN, GROUP_BY,
COMBINER
job_1789136053647_0019  1  1  2  2  2  2  2  2  2  final_top10_played_by_rating,grp_for_ratings,sorted,top10_played_ratings  G
ROUP_BY /steam/pig/Top10PlayedByRating,

Input(s):
Successfully read 50873 records (4864154 bytes) from: "/steam/games.csv"
Successfully read 41154795 records (2023286021 bytes) from: "/steam/recommendations.csv"

Output(s):
Successfully stored 87 records (4687 bytes) in: "/steam/pig/Top10PlayedByRating"

Counters:
Total records written : 87
Total bytes written : 4687
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789136053647_0017 -> job_1789136053647_0018,
job_1789136053647_0018 -> job_1789136053647_0019,
job_1789136053647_0019

2026-09-11 20:46:57,464 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:46:57,476 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:46:57,541 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:46:57,553 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:46:57,623 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:46:57,638 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:46:57,738 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:46:57,752 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:46:57,832 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:46:57,840 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:46:57,916 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:46:57,930 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:46:58,052 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:46:58,117 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:46:58,126 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:46:58,189 [main] INFO org.apache.hadoop.yarn.client.DefaultNoHARMFailoverProxyProvider - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 20:46:58,202 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 20:46:58,268 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 7 time
(s).
2026-09-11 20:46:58,268 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-11 20:46:58,390 [main] INFO org.apache.pig.Main - Pig script completed in 4 minutes, 3 seconds and 280 milliseconds (243280 ms)
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$

```

Figure 32: End of the run: 87 records stored, in 4 minutes 3 seconds.

Output

```

rii@pop-os: ~/Steam-Data-Analysis-using-Hadoop
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$ hdfs dfs -cat /steam/pig/Top10PlayedByRating/part-2
2026-09-11 20:46:59,072 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java clas
ses where applicable
Mixed 578080 PUBG: BATTLEGROUNDS 97733 30670189.2
Mixed 304390 FOR HONOR™ 91868 16358308.1
Mixed 1938090 Call of Duty® 133275 15254506.1
Mixed 1468810 亮谷八元 Tale of Immortal 116568 11724340.3
Mixed 1517290 Battlefield™ 2042 106386 9027781.1
Mixed 1418630 Dread Hunger 30806 8685620.5
Mixed 1142710 Total War: WARHAMMER III 35506 7251795.4
Mixed 1811260 EA SPORTS™ FIFA 23 37294 6646433.0
Mixed 433850 Z1 Battle Royale 36444 6645658.0
Mixed 680420 OUTRIDERS 45398 4166422.1
Negative 315330 All Guns On Deck 322 3178.9
Negative 719120 O'Fox Life 105 1709.2
Negative 388160 Project Druid - 2D Labyrinth Explorer- 96 1702.2
Negative 301220 Legends of Persia 103 1080.5
Negative 265240 Crazy Machines: Golden Gears 23 928.6
Negative 285090 Robowars 199 891.0
Negative 444420 24 Hours 'til Rescue 66 593.1
Negative 358700 3d Bridges 15 541.9
Negative 382160 Dark Years 75 423.5
Negative 397580 Moonbase 332 100 419.6
Positive 17510 Age of Chivalry 1019 45333.1
Positive 17700 INSURGENCY: Modern Infantry Combat 1052 39599.6
Positive 1190990 Colormeleons 49 12397.2
Positive 769920 Odysseus Kosmos and his Robot Quest: Episode 1 288 10527.9
Positive 1697460 Coin Pusher World 23 6452.2
Positive 640890 Will Glow the Wisp 732 5110.6
Positive 2091130 True Idle Simulator 31 4967.5
Positive 339160 Tales of the Orient: The Rising Sun 246 4862.1
Positive 376770 Big Thinkers 1st Grade 30 4708.4
Positive 1815370 咸鱼挂机游戏 17 4367.7
Very Negative 252910 Skyscraper Simulator 235 1357.2
Very Negative 273770 Game Tycoon 1.5 379 1039.1
Very Negative 246940 Lords of the Black Sun 102 948.0
Very Negative 273030 Pro Rugby Manager 2015 26 755.4
Very Negative 422420 Godus Wars 211 730.7
Very Negative 238870 Citadels 129 630.5
Very Negative 369080 Age of Survival 136 622.8
Very Negative 260510 World Basketball Tycoon 38 498.9
Very Negative 259890 Agricultural Simulator: Historical Farming 139 380.5
Very Negative 374640 March of Industry: Very Capitalist Factory Simulator Entertainments 40 345.0
Very Positive 440 Team Fortress 2 319492 101809977.6
Very Positive 730 Counter-Strike: Global Offensive 219737 94260307.3
Very Positive 570 Dota 2 216914 93072785.4
Very Positive 252490 Rust 278684 92944084.6
Very Positive 359550 Tom Clancy's Rainbow Six® Siege 189603 68445823.3
Very Positive 1085660 Destiny 2 185527 62504172.8
Very Positive 394360 Hearts of Iron IV 143187 53086631.1
Very Positive 346110 ARK: Survival Evolved 159736 46720359.4
Very Positive 230410 Warframe 127812 46450175.3
Very Positive 271590 Grand Theft Auto V 163909 46170394.5
Mostly Negative 1665460 eFootball™ 2024 24316 1407345.3
Mostly Negative 1504570 Cultivation Tales 7368 810267.0
Mostly Negative 1832640 Mirror 2: Project X 59606 635685.7
Mostly Negative 730310 DYNASTY WARRIORS 9 4592 205546.0
Mostly Negative 221020 Towns 1128 56059.8
Mostly Negative 282560 RollerCoaster Tycoon World™ 1991 32321.2
Mostly Negative 313010 Cities XXL 800 24712.7
Mostly Negative 238240 Edge of Space 1251 23510.2
Mostly Negative 47400 Stronghold 3 Gold 1487 21309.8
Mostly Negative 247950 Sacred 3 1624 16881.5
Mostly Positive 1599340 Lost Ark 129522 27429692.8
Mostly Positive 236390 War Thunder 80609 25339867.3
Mostly Positive 221100 DayZ 94365 24617576.3
Mostly Positive 438100 VRChat 98575 22353869.0
Mostly Positive 275850 No Man's Sky 171816 22017233.7
Mostly Positive 1063730 New World 96549 20351309.2
Mostly Positive 386360 SMITE® 64689 18246201.5
Mostly Positive 359320 Elite Dangerous 68123 17392171.5
Mostly Positive 440900 Conan Exiles 49671 12815929.5
Mostly Positive 552900 World of Warships 56992 12475012.6
Overwhelmingly Negative 1180320 War of the Three Kingdoms 11274 255526.1
Overwhelmingly Negative 246090 Spacebase DF-9 1782 26679.2
Overwhelmingly Negative 201510 Flatout 3: Chaos & Destruction 2010 9768.2
Overwhelmingly Negative 227160 Kinetic Void 660 7065.4
Overwhelmingly Negative 267600 Airport Simulator 2014 466 4775.4
Overwhelmingly Negative 1563130 SuperPower 3 480 2285.9
Overwhelmingly Negative 1799220 Malice 495 1702.5
Overwhelmingly Positive 4000 Garry's Mod 137199 41697656.1
Overwhelmingly Positive 105600 Terraria 123071 35443053.9
Overwhelmingly Positive 250900 The Binding of Isaac: Rebirth 114002 33914457.8
Overwhelmingly Positive 294100 RimWorld 100999 33354178.5
Overwhelmingly Positive 227300 Euro Truck Simulator 2 179407 32000932.1
Overwhelmingly Positive 292030 The Witcher® 3: Wild Hunt 204166 31522739.6
Overwhelmingly Positive 427520 Factorio 112195 28700677.3
Overwhelmingly Positive 550 Left 4 Dead 2 179684 24630056.2
Overwhelmingly Positive 284160 BeamNG.drive 127258 23911526.3
Overwhelmingly Positive 49520 Borderlands 2 158945 22867857.1
rii@pop-os:~/Steam-Data-Analysis-using-Hadoop$

```

Figure 33: Top 10 most played games in every rating group.

The script stored 87 rows, not 90. The Overwhelmingly Negative group has 14 games, but only 7 of them have any review in recommendations.csv, so that group can only return seven.

Visualization

Top 10 games by total hours played in each rating group

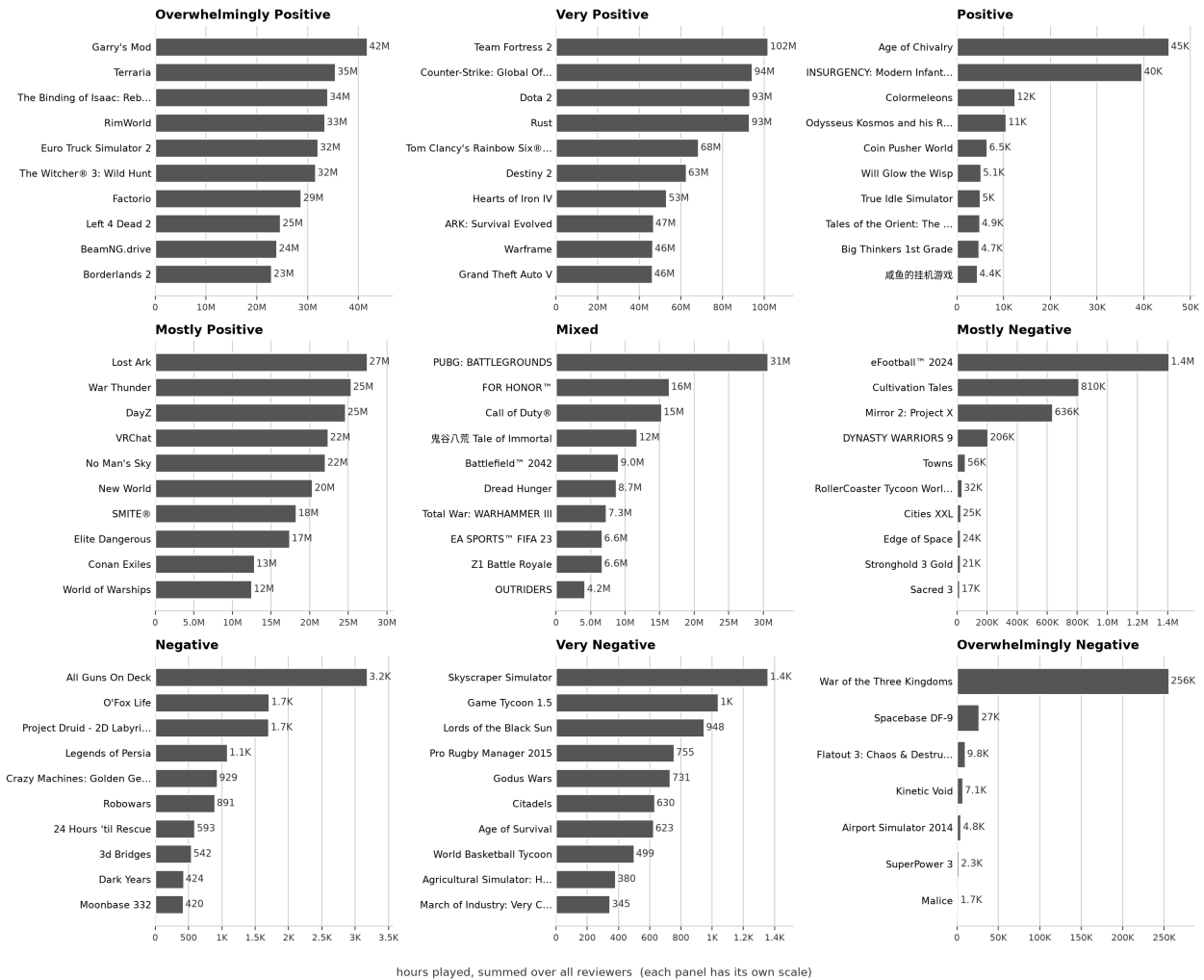


Figure 34: Top 10 games by total hours played in each rating group. Each panel has its own scale.

7. Monitoring the Jobs

Hadoop's web interfaces were kept open while the jobs ran.

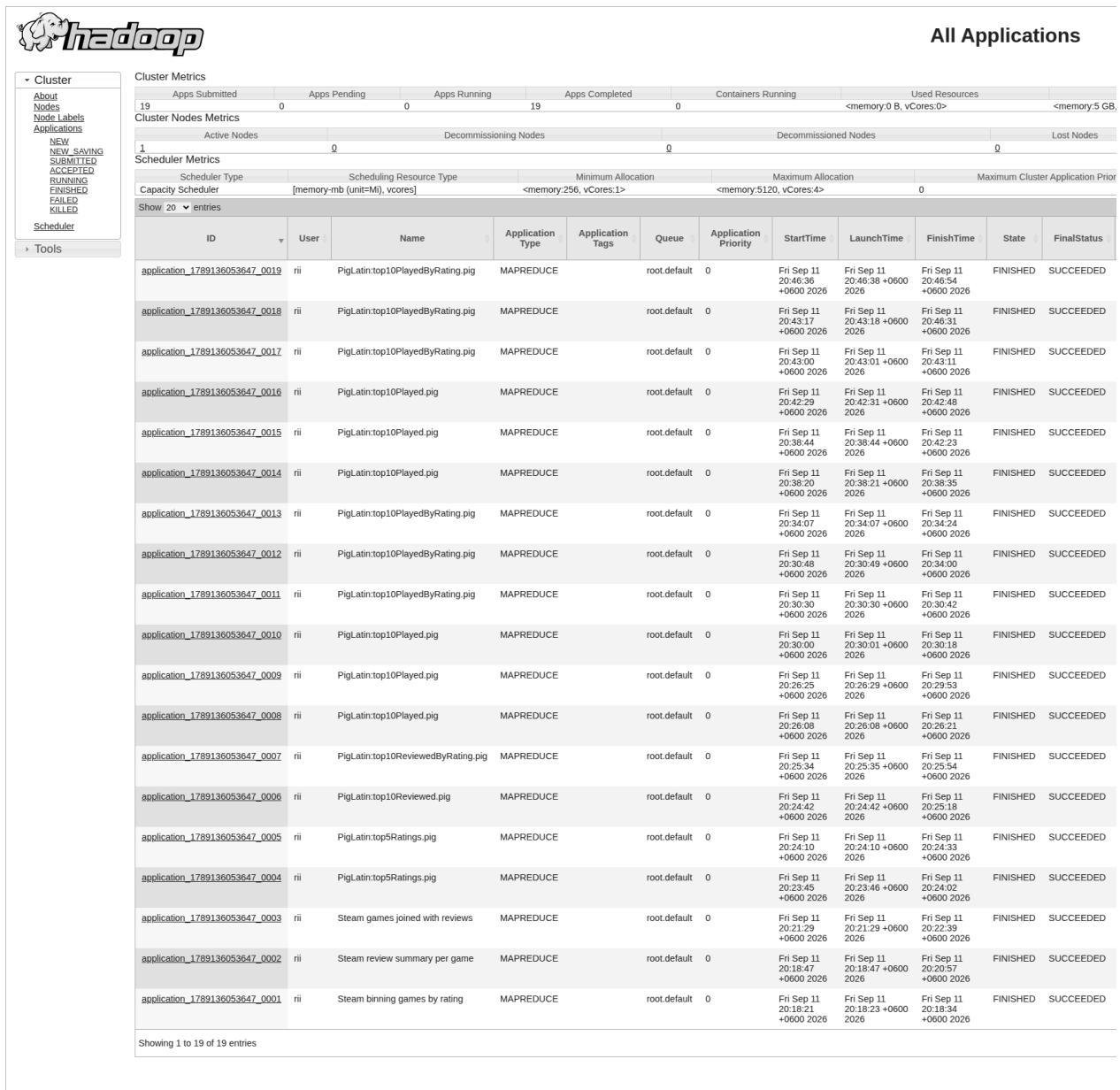


Figure 35: YARN ResourceManager (localhost:8088): all 19 applications finished with status SUCCEEDED.

The 19 applications are the three MapReduce jobs and the MapReduce jobs Pig generated for the five scripts. Scripts 4 and 5 appear twice: they were run once, then run again after changing only how the hours are printed (from Pig's default scientific notation to one decimal place). The second runs are the ones in this report.

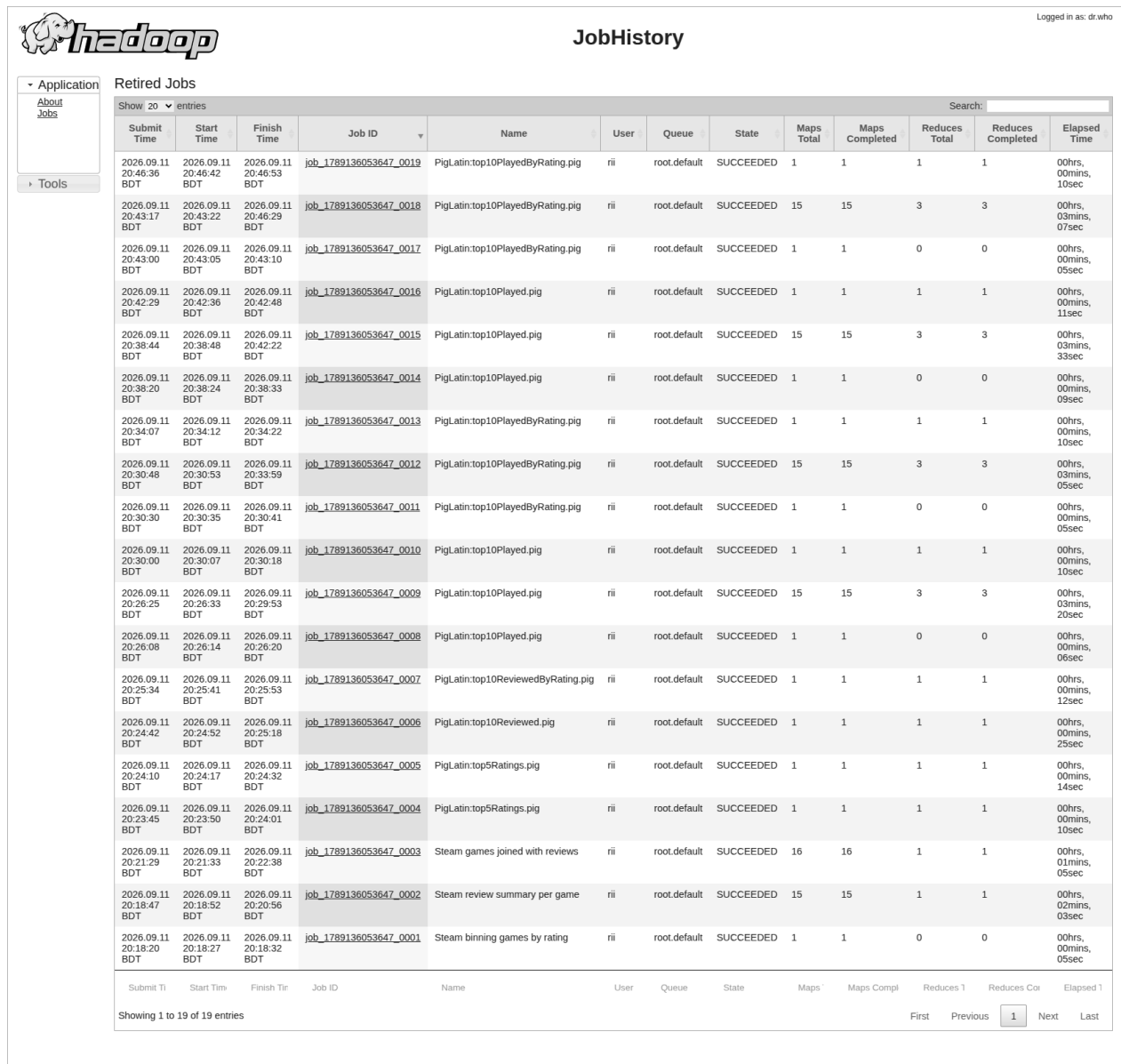


Figure 36: MapReduce JobHistory server (localhost:19888) with the map and reduce task counts of every job.

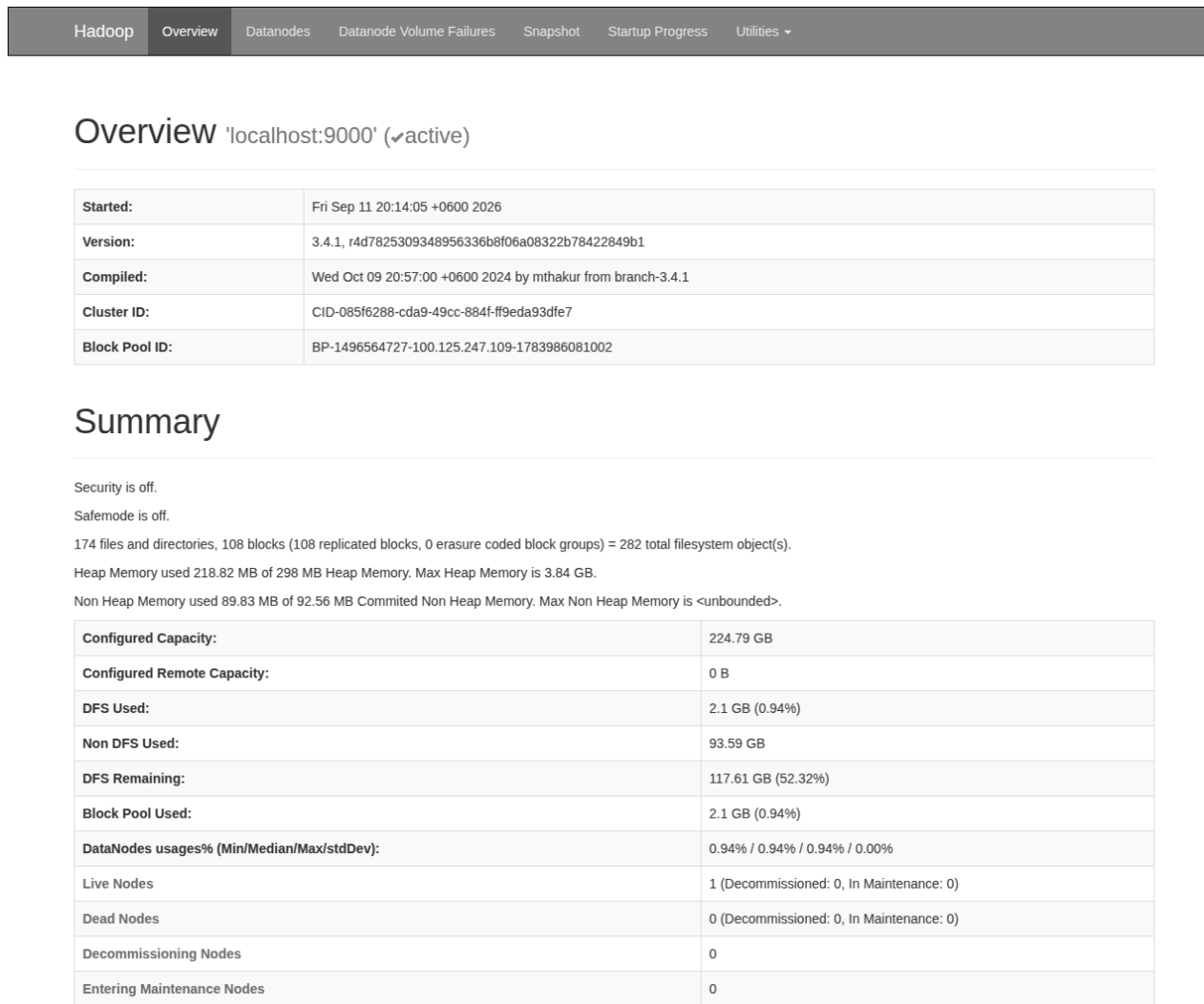


Figure 37: NameNode overview (localhost:9870).

Hadoop

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[Snapshot](#)
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Browse Directory

📁

📄

📄

📄

Show 25 entries
Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	
☐	-rw-r--r--	rii	supergroup	4.64 MB	Sep 11 20:15	1	128 MB	games.csv	🗑
☐	drwxr-xr-x	rii	supergroup	0 B	Sep 11 20:18	0	0 B	out_binning	🗑
☐	drwxr-xr-x	rii	supergroup	0 B	Sep 11 20:22	0	0 B	out_join	🗑
☐	drwxr-xr-x	rii	supergroup	0 B	Sep 11 20:20	0	0 B	out_summary	🗑
☐	drwxr-xr-x	rii	supergroup	0 B	Sep 11 20:46	0	0 B	pig	🗑
☐	-rw-r--r--	rii	supergroup	1.88 GB	Sep 11 20:15	1	128 MB	recommendations.csv	🗑
☐	-rw-r--r--	rii	supergroup	184.12 MB	Sep 11 20:15	1	128 MB	users.csv	🗑

Showing 1 to 7 of 7 entries

Previous

1

Next

Hadoop, 2024.

Figure 38: The /steam directory in the HDFS browser: the three input files and the output directories of every job.

8. Verification

Several results can be checked against each other or against the raw file, because different jobs reach the same number by different routes. All checks passed.

Check	Expected	Produced	
Reviews read by the summary job = rows of recommendations.csv	41,154,794	41,154,794	pass
Reviews summed over the summary output = over the join output	41,154,794	41,154,794	pass
Games in the summary output = games matched by the join	37,610	37,610	pass
Joined games + games with no reviews = rows of games.csv	50,872	37,610 + 13,262	pass
Nine bins of the binning job add up to all games	50,872	50,872	pass
Bin sizes = counts computed directly from games.csv	9 values	all 9 equal	pass
Pig top 5 rating groups = the five largest bins	5 values	all 5 equal	pass
Pig top 10 by hours = MapReduce join top 10 (ids, reviews, hours)	10 rows	identical	pass
Pig-5 rows = 8 groups $\times$ 10 + 7 reviewed Overwhelmingly Negative games	87	87	pass

Table 6: Cross-checks between jobs.

The bin sizes were also counted outside Hadoop, straight from the CSV, before any job was run, and the Pig-versus-MapReduce comparison of the top ten was done with diff on the two outputs.

9. What the Data Says

Steam reviews are mostly positive. 85.8% of the 41 million reviews recommend the game, and 71.7% of all games carry one of the four positive labels. Only 2,226 games (4.4%) carry a negative one, and just 14 are Overwhelmingly Negative.

Attention is concentrated in a few games. Of the 50,872 games, 37,610 appear in the review data at all, and half of those have 39 reviews or fewer. The ten most reviewed games — 0.03% of the reviewed games — hold 5.4% of all reviews.

The most played games are long-running multiplayer titles. Team Fortress 2, Counter-Strike: Global Offensive, Dota 2 and Rust each passed 90 million hours of play among their reviewers, and the average Counter-Strike or Dota 2 reviewer had played about 429 hours. Across all reviews the players had logged 4.14 billion hours.

Popular does not mean well rated. PUBG: BATTLEGROUNDS is the second most reviewed game on Steam but is rated Mixed, with 57% positive reviews; Overwatch® 2 has 181,198 reviews and is Overwhelmingly Negative. At the other end, Wallpaper Engine is 98.2% positive in the review data.

A rating label is partly a volume label. The Positive and Negative groups top out at 49 reviews, because Steam reserves the “Very” and “Overwhelmingly” labels for games with more reviews. Grouping by rating therefore mixes two things — how good the reviews are and how many there are — which is worth keeping in mind before comparing the groups directly.

10. Conclusion

A 2.2 GB, three-table Steam dataset was loaded into HDFS and analysed on a single-node Hadoop 3.4.1 cluster with three Java MapReduce programs and five Pig Latin scripts, all of which completed successfully.

The three MapReduce programs each show a different design pattern. The summarization job shows how a custom Writable and a combiner shrink 41 million map outputs to under a hundred thousand before the

shuffle. The binning job shows a map-only job writing many outputs at once. The reduce-side join shows two differently shaped inputs meeting in one reducer, and how in-mapper combining makes the join over 41 million rows finish in just over a minute.

Pig answered the ranking questions in 15 to 33 lines per script, including a replicated join over the full review table, but took roughly three to four times as long as the hand-tuned Java join for the same answer. The two tools agreed on every result they share, which is the strongest evidence that both are correct.

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